

STarlight Acquisition and Reflection towards Interferometry (STARI) CubeSat Design Group 4

Critical Design Review



MANE 4250 - Space Vehicle Design
Rensselaer Polytechnic Institute

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List of Acronyms

ADCS	Attitude Determination and Control System
CDR	Critical Design Review
EKF	Extended Kalman Filter
FPS	Electric Power System
FIIV	Extreme Ultraviolet
FCC	Federal Communications Committee
FRAM	Ferroelectric Random Access Memory
DoF	Degree of Freedom
GEO	Geostationary Orbit
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
LEO	Low Earth Orbit
NASA	National Aeronautics and Space Administration
OBC	On-Board Computer
PCB	Printed Circuit Board
PID	Proportional-Integral-Derivative controller
RAM	Random Access Memory
SSP	Sub-Satellite Point
SDRAM	Synchronous Dynamic Random Access Memory
SSO	Sun Synchronous Orbit
STARI	Starlight Acquisition and Reflection towards Interferometry
STK	Satellite Tool Kit
TRL	Technology Readiness Level
UHF	Ultra-High Frequency
VHF	Very High Frequency
VISORS	Virtual Super Optics Reconfigurable Swarm

1 Introduction (M. Bakken)

Interferometry is a powerful technique used in astronomy and other scientific fields to measure and analyze the interference patterns of waves, such as light, radio, and sound. In its simplest form, interferometry works by splitting a beam of light into two separate paths. These beams travel different distances and are then recombined to create an interference pattern. This pattern provides detailed information about the waves' phase differences and can reveal properties of the light source, such as its position, shape, and structure [1].

In astronomy, interferometry is used to improve the angular resolution of telescopes, the ability to distinguish between two closely spaced objects in the sky. Typically, the resolution of a telescope is limited by its aperture, or the size of its mirror. Larger apertures allow for finer resolution, but building extremely large ground-based telescopes is challenging and expensive. Interferometry overcomes this limitation by combining the light from multiple smaller telescopes spread out over a large area. This effectively simulates a telescope with a much larger aperture, allowing astronomers to achieve the same resolution as a giant telescope, even though the individual telescopes are much smaller.

On Earth, telescopes are often restricted in what can be seen. This is due to atmospheric disruptions along with the angular resolution of a telescope. Angular resolution is the smallest angle between objects that can be seen clearly as separate. For seeing celestial bodies far beyond the solar system, increasing angular resolution is imperative. Interferometry accomplishes this by having multiple satellites reflecting starlight to create a more defined picture. The first optical interferometer was used to measure the diameter of the red giant star Betelgeuse [2]. Earth-based interferometry can still run into similar problems as a normal telescope would such as atmospheric distribution and geography.

Space-based interferometry offers a solution to these limitations. Without Earth's atmosphere blur incoming light and without the physical constraints of the ground, space-based telescopes can achieve far greater angular resolutions. This is critical for tasks such as imaging exoplanets, studying the accretion disks around black holes, and probing the atmospheres of distant planets for potential biomarkers, molecules that may indicate the presence of life.

Recent missions have been utilizing these techniques, namely the VISORS mission, which was designed to gather high resolution images of the sun in the EUV [3]. This was done by using two CubeSats with one being the optics spacecraft and the other being the detector spacecraft. The optics spacecraft hosts a photon sieve, which functions as a high-resolution lens that captures and focuses the EUV light. The focused light is then transmitted to the detector spacecraft, where the image is recorded. This mission's goal is to study the Sun's corona in unprecedented detail, enhancing the understanding of solar phenomena such as coronal heating and solar wind formation.

The VISORS mission has been an invaluable resource for the development of STARI due to its unique approach to formation flying and its successful demonstration of key technologies that STARI will build upon such as transmission of data between two satellites. Both VISORS and STARI involve CubeSat platforms, which are small, cost-effective satellites that are ideal for testing advanced space technologies. VISORS, in particular, has laid the groundwork for space-based interferometry by addressing many of the same technical challenges that STARI aims to

overcome. Although STARI will not work as an interferometer, its success will pave the way for future space interferometry missions that could transform the understanding of the universe.

2 Mission Overview

2.1 Mission Objective (N. Jankowski)

The objective of STARI is to develop system-level technologies for transmitting starlight between satellites, demonstrating the plausibility of precise formation flying in orbit. To accomplish this, a pair of satellites equipped with identical payloads composed of a transmitter and receiver will be placed into orbit. Nicknamed STARI-1 and STARI-2, the two spacecraft will redirect starlight between one another despite being tens to hundreds of meters apart.

With the broad goal stated, the specific objectives of the STARI mission can now be defined. With this mission being the first of its kind, the primary focus is validating the formation flying approach and raising system/subsystem TRLs. The objective and its corresponding priority level are summarized in Table 2.1.1.

Table 2.1.1 - STARI Mission Objectives

Objective	Priority
Demonstrate successful reflection from one free-flying spacecraft to another.	1
Demonstrate stable injection into single-mode fiber continuously for at least 10 seconds.	2
Passively safe orbit management of 2-spacecraft flying formation.	3
Explore system performance for spacecraft separation from 10 to more than 100 meters.	4
Optimize fiber injection algorithms for stable and high coupling efficiency.	5

2.2 Mission Requirements (J. Mitchell)

To fulfill the objectives of the mission, the mission has requirements for each subsystem that must be fulfilled, ensuring the mission is successful. These main requirements were identified and listed below:

- Attitude must be controllable within 2 degrees of the desired angle between the two CubeSats over 100 meters for a stable injection into single-mode fiber
- Position determination must be accurate within 10 cm of the actual position of the spacecraft using a combination of GPS and other sensors
- The reference lasers in the payload require an angular accuracy of 2 degrees over a distance of 100 meters.
- The CubeSat must be able to house a 1Ux1Ux2U payload weighing 3kg and consuming an idle power of 1.2W and 20W during experimentation
- The payload must be kept between -40°C and 90°C during operation
- Stars used during experiment must have a magnitude of 3 or higher
- The CubeSat must be able to handle data coming from the payload at a rate of 10 kBps
- The mission must demonstrate fulfill its objectives within 1 year

2.3 Major Mission Risks (J. Mitchell)

There are mission scoped risks associated with the mission that must be considered to ensure mission success mainly dealing with communication between the two spacecraft, and the interactions between subsystems onboard the spacecraft. These risks were identified and mitigation strategies for each were developed.

Risk	Mitigation
Failure of components within one of the payloads	Both CubeSats have identical payload modules. If a sensor or actuator within one of the payload modules stops working (one cannot transmit data), the other payload can take over.
One of the spacecrafts completely loses ground communication	Because the satellites are close to each other in orbit, inter satellite communication can be used to diagnose the issue, or carry out the rest of the mission.
One of the spacecrafts is uncontactable or lost	Objectives involving starlight transfer between the two cannot be completed, but objectives involving precision control of the spacecraft will still be carried out.
Objectives not met; mission exceeds expected length of operation	Decommissioning will be delayed via ground command until mission objectives are met, or until system failure occurs such as running out of propellant.

2.4 Technical Constraints (J. Mitchell)

2.4.1 Time Constraints

The STARI mission does not have a fixed operational timeframe; instead, the duration in orbit will be determined by the successful demonstration of all mission requirements. The time needed to validate technologies, such as starlight reflection and formation flying, will influence the overall timeline. The goal is to aim for a minimum 2-year operation time as it will allow for a sufficient number of observations while demonstrating formation flying ability. This will allow for the opportunity to view numerous stars throughout the course of the mission, which would not be possible with a shorter operation time. Ultimately, the mission will remain in orbit until all objectives are met and validated comprehensively.

2.4.2 Launch Vehicle Constraints

The satellite will be placed into orbit with other satellites through a rideshare program with a launch vehicle provider. For this mission, the selected launch vehicle is SpaceX's Falcon 9. As outlined in the Rideshare Payload User's Guide [29], all payloads launched on a SpaceX Rideshare Mission must undergo either flight unit protoflight environmental verification testing (if adopting a protoflight qualification approach) or acceptance verification testing (if utilizing a unit/fleet qualification and acceptance approach) at the fully integrated payload assembly level. This requirement ensures that the separation from the launch vehicle occurs successfully and that the payload is protected from potential damage caused by vibrations during launch.

Additional considerations include the following: the dispenser holding the CubeSat must maintain a center of gravity that is not more than 600 mm offset from the payload origin according to the SpaceX Rideshare Payload User's Guide. The payload dispenser must comply with the design specifications of the CubeSat plate provided by SpaceX. The payload is required to achieve a minimum separation velocity of 0.3 m/s. Attitude maneuvers and deployment of components must be postponed for at least 120 seconds after separation, while propulsion maneuvers must be delayed for a minimum of 7 days post-separation. Additionally, engineers are required to comply with the guidelines set forth in the AFSPCMAN 91-710 Range User's Manual for the design and operation of both flight and ground systems.

2.5 Non-Technical Considerations

2.5.1 Environmental Considerations (M. Bakken)

Environmental considerations are a critical factor in the development and planning of the mission. One consideration is that of pollution, both in space and on Earth. This is especially important with the rise of climate change. The Earth pollutants will be taken into consideration by choosing more environmentally friendly propellants or manufacturing sources. Furthermore, on Earth, there are issues of noise pollution causing trouble for certain species, such as migrating birds [33]. It is important to choose an appropriate launch location to not disturb the natural environment.

In space, it is paramount to avoid creating more space debris. To do this, careful planning on decommissioning the satellite must be done. This will be done by utilizing a drag sail in order to decommission the CubeSats in under a year after operation. Space also has a very competitive orbit availability, specifically in Low-Earth orbits or geosynchronous orbits. The spacecraft must make great considerations to not collide or interfere with these satellites.

Additionally, the environmental impact of the mission must be addressed, particularly regarding the emissions and fuel use of the launch vehicle. Space launches contribute to atmospheric pollution, and the growing frequency of such launches raises concerns about their cumulative environmental footprint. There is an ethical responsibility to consider sustainable practices, such as offsetting carbon emissions or exploring more eco-friendly propellants. Transparency throughout the mission is also crucial, both in terms of the technology used and the mission's objectives. Open communication with the public and the scientific community ensures accountability and builds trust.

2.5.2 Political and Legal Considerations (A. Cohen)

Because the satellite will interact with many political and legal entities, many possible considerations must be analyzed to ensure the craft follows all necessary laws. According to Article 7 of the Outer Space Treaty, the craft is subject to the laws of the state from which it was launched. In the United States, requirements and regulations are listed in the CFR [34]. For the satellite to communicate through radio transmission a license must be obtained from the FCC allowing the craft use of a band of radio spectrum in compliance with the radio regulations of the International Telecommunications Union. Orbital elements must be provided to NASA and meet its Tracking and Data Relay Satellite operational requirements to allow tracking from the Flight Dynamics Facility [35]. The disposal of the satellite at the end of its lifetime must adhere to the space debris guidelines of the Inter-Agency Space Debris Coordination Committee [36].

2.5.3 Cost Considerations (J. Mitchell)

Cost management is a critical component in the development and operation of any mission. For this particular mission, the total allocated budget is \$10 million, which must cover all expenses related to the construction of two CubeSats housing the STARI payloads. A

comprehensive identification of all potential expenses ranging from payload development to launch and maintenance is essential to ensure the mission's success.

The development of the payload will incur several costs, including the procurement of individual components, shipping and delivery fees, and software licensing throughout the design and construction phases.

Several mission-specific fees must also be accounted for, particularly those related to the satellite launch, regulatory licensing, and communications infrastructure. For instance, federal licensing fees from agencies such as the FAA and FCC will be required. The FCC licensing fee for a small, non-geostationary satellite is approximately \$12,215 per license [36]. While the precise FAA licensing fee for orbital missions is not available, it is likely to be required depending on the mission's trajectory and launch parameters. Rideshare launch services, such as SpaceX, charge around \$300,000 for launching a satellite under 50 kg into a Low-Earth or Sun-Synchronous Orbit. [29] Additional integration fees may apply, covering elements such as adapters, cabling, and other required components for rideshare compatibility. Insurance costs are assumed to be covered by the rideshare provider and will not be treated as a separate budget item.

Ground operations, specifically uplink and downlink communications with satellites, must also be considered. The costs for these operations vary depending on the chosen solution. If a dedicated ground station is to be built, development costs and FCC licensing fees, which are approximately \$575, should be included in the budget [36]. Alternatively, if a third-party ground station service, such as Amazon's AWS Ground Station as a Service, is used, the associated fees will need to be determined and accounted for accordingly. By thoroughly evaluating these costs, the mission can better ensure the financial feasibility and overall success of its operations.

2.5.4 Ethical Considerations (V. Bindal)

One of the greatest ethical concerns in space missions is the contribution to space debris. The STARI project must take great care to avoid adding to the already significant debris problem in Earth's orbit. Mitigating such an issue involves planning for the decommission of both satellites, ensuring they are burned up in the atmosphere using the drag sail. This also has to follow strict adherence to international space debris mitigation guidelines, minimizing the long-term impact on the orbital environment and preserving space for future missions.

Another critical consideration is collision avoidance. With CubeSats flying in close formation and navigating potentially crowded orbits, the risk of accidental collisions with each other or other orbiting objects is a real issue. Having a robust ADCS system to ensure high positional accuracy will ensure the prevention of both CubeSats colliding. Ethical mission planning must include robust collision avoidance strategies, using an intensive array of sensors and actuators. Failing to address these risks adequately could not only jeopardize the STARI mission but also endanger other satellites and space infrastructure, making it critical to incorporate state-of-the-art ADCS systems

2.6 Design Changes from PDR

The largest change from the PDR involves experiment mode. Before, the plan was to have both satellites transmit and receive starlight at the same time. However, as the model was created, the only way for starlight to enter the vehicle was through the 3Ux1U side of the chassis. By doing this on both satellites, the transmitters of both would be aligned, rather than a transmitter/receiver pair. Hence, this change is required. Further explanation and figures are shown in Section 3.1.3. This allows the mission to have redundant payload function in the event of a component failure and simplifies the required angular and position control.

A second change that was implemented was the switch from lithium-polymer based battery material to a lithium-iron phosphate material. The power schedule included far more cycling of the battery than anticipated, warranting this change, and this is discussed further in the Power section of the CDR.

The third change was that the ΔV budget for formation activity was updated from the PDR. This was due to the choosing a formation that more accurately suited the mission objectives. The change was defined by using the formation simulation's ability to accurately predict the necessary ΔV in order to maintain the chosen formation. This will be discussed in more depth further into the CDR.

3 Subsystems

3.1 Structure (N. Jankowski)

3.1.1 Structure Analysis Overview

To finalize the structure of the STARI CubeSats, specific components and their locations must be determined. To begin this process, each component from every subsystem will be listed with their mass and volume. The reasoning for this is to include as much detail about the entire system as possible and demonstrate that every part is accounted for when creating the CAD model. This information will be assembled in a simple table.

The second step in finalizing the structure is to develop a 3D model using Siemens NX that organizes every component onto the chassis using an optimal layout. This layout will be determined by understanding the requirements of each subsystem, specifically ADCS. Understanding the moment of inertia and center of mass (CoM) of the satellite at various stages is important information to feed to the ADCS system for determining attitude control. Two models will be created for pre and post solar panel deployment, which will have a large impact on inertia.

In a next iteration, the impact of propellant levels should be accounted for, measuring the change in center of mass and inertia as the tanks empty. All of these changes should be assembled in a time plot, showing the differences in all 3 axes as the mission progresses. Doing so ensures that the ADCS system has a full understanding of the changes occurring to the CubeSat and will not overestimate attitude control.

3.1.2 Mass and Volume Budget

To begin the organization of the STARI satellites, a final mass and volume budget will be established. This will be much more in-depth than the budgets displayed in the PDR, and will list every component of each subsystem. By doing so, the budget will act as a checklist for the CAD model, ensuring that every component is represented accurately. Additionally, location considerations will be listed for each component. In designing a CubeSat, location plays a massive role in the success and efficiency of its subsystems. Where a component is located can have major impacts on inertia, center of mass, thermal environment, and accuracy of measurements. All of this data is tabulated in Table 3.1.1.

Table 3.1.1 - Finalized Mass/Volume Budget [17]

Subsystem	Component	Mass (kg)	Volume (cm ³)	Considerations
Structure	Chassis	0.674	NA	
Power	Solar Panel Array	1.31	NA	XY vs YX deployment
	EPS	1.423	828.7	
ADCS	iADCS400	1.15	615.718	Star tracker needs to be pointed outwards.
	Pulse-40	0.012	11.172	Needs to be as close to CoM as possible (sweet spot as CoM shifts)
	AsteRx-m3 Pro+	0.02	30.989	
Commands & Data	iOBC	0.094	107.136	
Telecommunications	Antenna	0.089	67.228	Placed in location with limited potential blockers of signal
	Transceiver	0.025	19.698	Placed near antenna
Propulsion	MiPS	0.848	360	Thrusters need to be pointed outward and in line with CoM to avoid torque
Mechanisms & Deployables	PICO Drag Sail	0.45	390	Deployment side must face outward
Thermal Management	Thermofoil Heaters	0.05	NA	Placed on thermally sensitive components
Payload	-	3.0	2270	Requires two outlets on the front face. Requires one inlet perpendicular to outlets
SUM		9.145	4700.641	

3.1.3 Satellite Layout

Using the components established in Section 3.1.2, a 3D model of the satellite can be created. When available, the vendor CAD models are utilized. In most cases, these models are restricted from use, so a simple model with appropriate dimensions and a scaled density to match the mass of the component properly is used instead. Scaled densities were calculated by taking

the actual mass as defined by vendor specifications and dividing it by the CAD volume of the component. This is needed as CAD models may not accurately capture the exact volume, but an exact mass is crucial in generating useful CoM and inertia data points. The heaters are not modeled currently and require thermal analysis to determine location. However, they are extremely low in mass and volume and would have little impact on the output values. The undeployed model of the satellite can be seen in Figure 3.9.1. Alterations to the shear panels of the chassis were necessary to accommodate the propulsion and drag sail components. However, these changes were minor and should not affect the structural integrity of the chassis. Additionally, the vendor (AAC Clyde Space) supports custom configurations. The solar panels also had to be shortened, removing the top tabs to ensure that the geometry stays within the limits as defined by CubeSat standards [4]. The vendor (EnduroSat) also supports custom configurations, so this alteration is not unrealistic.

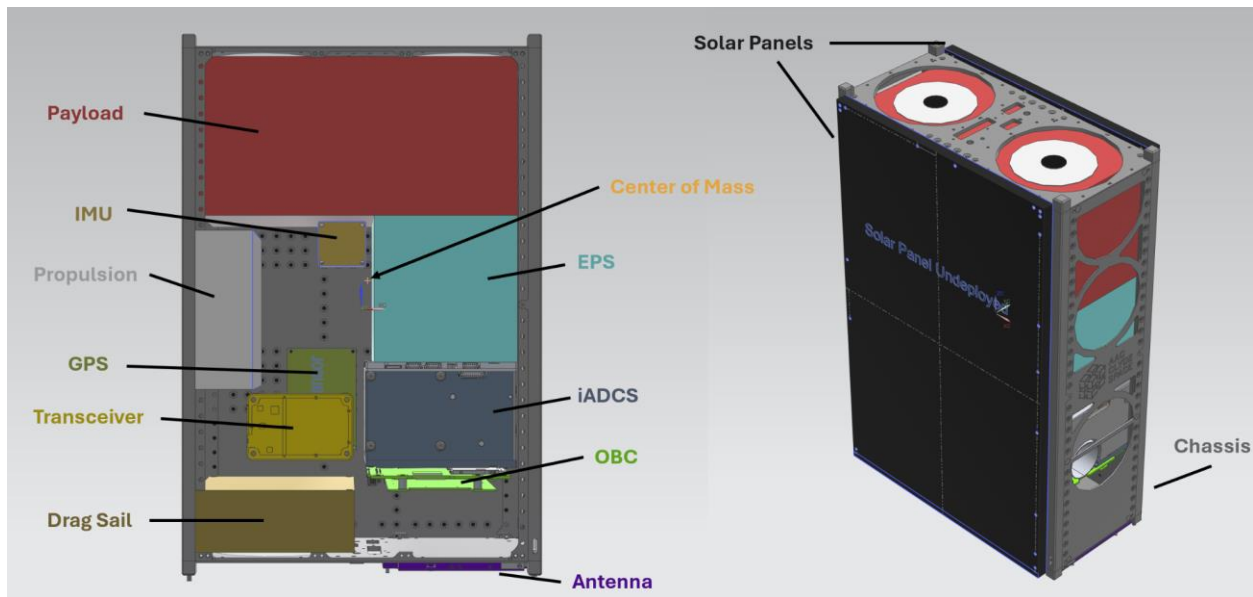


Figure 3.1.1 - Undeployed Satellite Configuration

Crucial characteristics of the satellite layout involve center of mass location and inertia matrix. The results are tabulated in 3.1.2. It is important to note that the reference coordinate system of these values is the geometric center of the satellite, which can be seen in Figure 3.9.1.

Table 3.1.2 - Undeployed Characteristics

Center of Mass (mm)	[4.1771, -0.0056, 19.0421]		
Inertia Matrix (kg*cm ²)	$\begin{vmatrix} 896.43 & 1.512 & -18.627 \\ 1.512 & 1147.35 & -1.405 \\ -18.627 & -1.405 & 465.76 \end{vmatrix}$		
I_{xx}	I_{xy}	I_{xz}	
I_{xy}	I_{yy}	I_{yz}	
I_{xz}	I_{yz}	I_{zz}	

Another model of the deployed configuration can be made, and a similar analysis of center of mass and inertia matrix can be conducted. The major deployables are the solar panels and the antenna. This inertia matrix is required for ADCS simulations and will be referenced in Section 3.3.

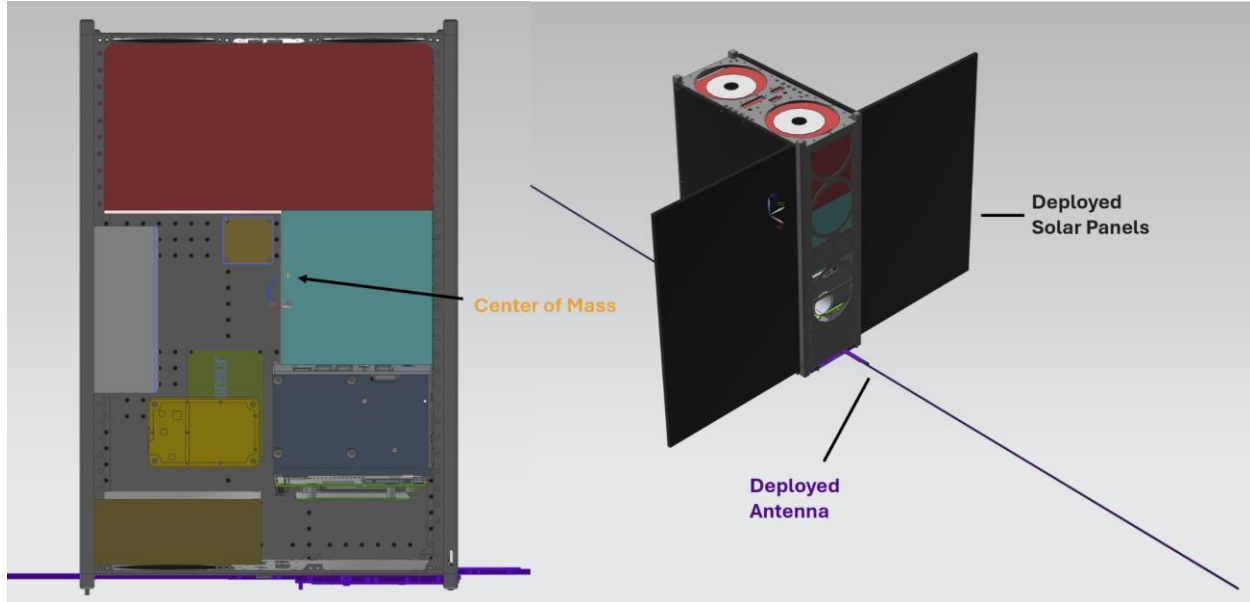


Figure 3.1.2 - Deployed Satellite Configuration

Table 3.1.3 - Deployed Characteristics

Center of Mass (mm)	[11.5778, 0.2156, 19.1781]		
Inertia Matrix (kg*cm²)			
	I_{xx}	I_{xy}	I_{xz}
	I_{xy}	I_{yy}	I_{yz}
	I_{xz}	I_{yz}	I_{zz}
	1062.86	2.928	-17.22
	2.928	1191.38	-3.489
	-17.22	-3.489	677.37

Between the two configurations, there is a clear change in moments of inertia, as the solar panel mass is expanded away from the center of mass. This causes a slight increase (~200 kg*cm²) in the XX, YY, and ZZ axes. This is not a negligible change, so positioning and thrust algorithms used in ADCS and propulsion must account for the shift in CoM and inertia in order to achieve proper attitude and velocity vectors.

In order for this mission to be successful, the structure must be able to accommodate all payload requirements. The payload requires an inlet for starlight acquisition, an outlet for starlight transfer, and an outlet for starlight receiving. The size of the solar panels makes this a challenge, as a larger portion of surface area is covered by them. In order to accomplish this, only one satellite may transmit starlight to the other at a time, so that the transmitter/receiver pair is aligned. The distance between the receiver and transmitter on the payload is 100 millimeters. Since the

payload is able to operate when the locating laser is within 2 degrees, this alignment can occur without translation. At 10 meters apart, the transmitter/receiver would be misaligned by 0.64° , well within the tolerance. In the event that the two satellites are out of tolerance due to various other sources of error, the propulsion system may be used to perform a purely directional, low-thrust maneuver before the experiment begins. The configuration for starlight acquisition can be seen in Figure 3.1.3.

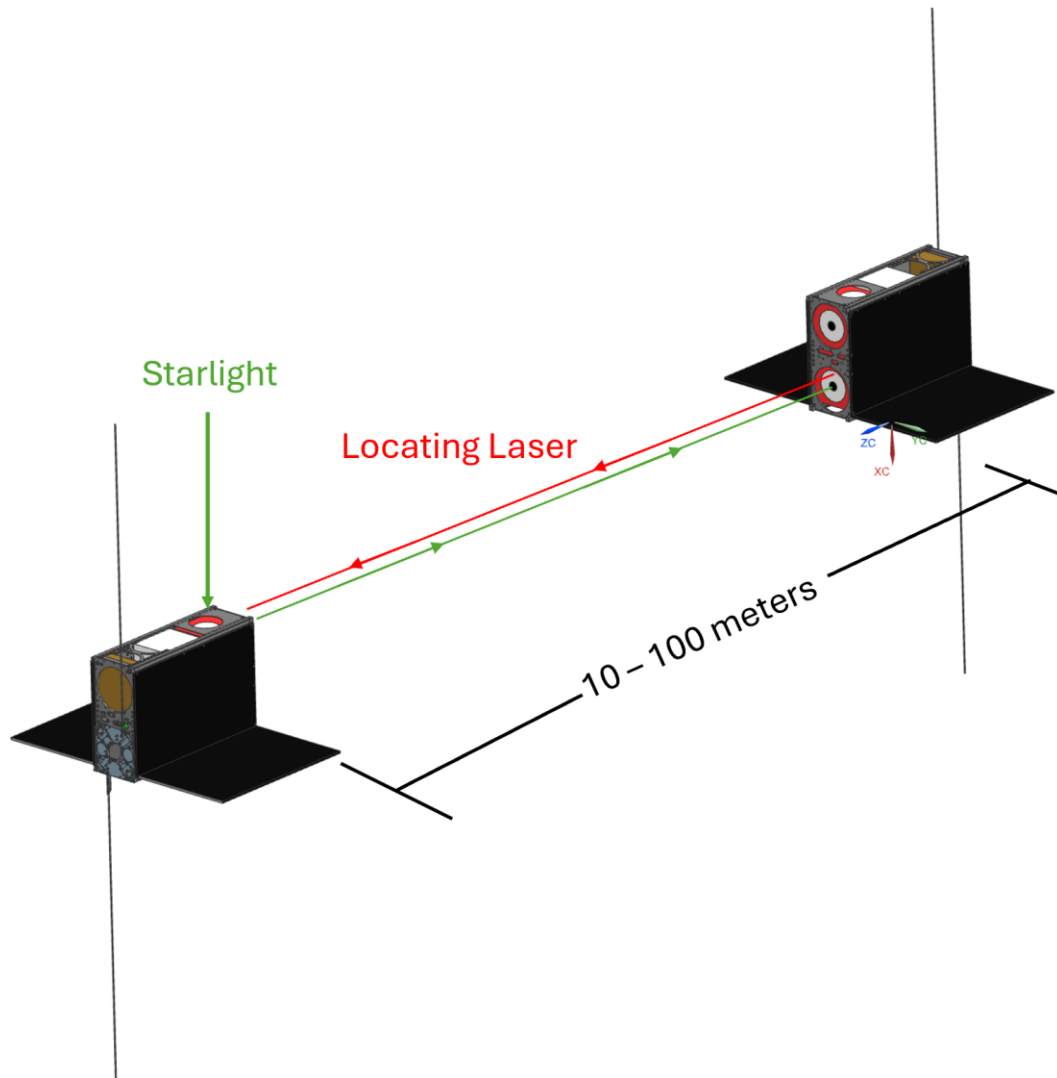


Figure 3.1.3 - Formation Flying Configuration

Starlight acquisition can be accomplished by either STARI satellite, as both are equipped with a transmitter and receiver pair in the event of component failure on either. This allows the satellite receiving starlight from the other to focus solely on keeping proper alignment to the transmitter, without having to stay precisely aligned with the star of interest. This simplifies each experiment and reduces the probability of falling out of the acceptable angular range.

3.1.4 Risk Assessment

Due to the nature of a CubeSat mission, its structure is rather risk-averse. The largest risk this subsystem holds is during its time on Earth, before it is launched into orbit. During assembly, test, and integration processes there is potential for damage to sensitive systems due to excessive forces. These could be caused by dropping the unit, banging it into something, or during transport from one site to another.

Less severe risks occur in orbit, primarily due to thermal reasons. As the satellite enters periods of lightness and darkness, the average temperature of the satellite and its components oscillate. Different components have different coefficients of thermal expansion, meaning abundant stresses can form if one part expands quicker than the rest.

Table 3.1.4 - Risk Mitigation Matrix

Risk ID/Title	Severity Score	Trend	Mitigation	Verification
001 Mishandling of System	High	↓	Ensure there are clear handling instructions (places safe to grab) and that the assembly process does not require excessive force.	All team members are trained appropriately for any handling and a form is signed that this training has been completed.
002 CTE Mismatch	Low	→	Ensure design has minimal mechanical overconstraint and components have similar expansion coefficients.	Accepted as is

3.2 Flight Mechanics (M. Bakken)

3.2.1 Overview

In order for the finalization of the flight mechanics system much capability and error must be considered. To determine the ability and capability of the STAR1 mission 3 key simulations will take place. These are the formation simulation, the total orbit simulation, and the decommissioning time. These simulations will also inform a more up to date and accurate ΔV budget that will be used in the propulsion system.

The Formation Simulation will ensure that the formation of the STAR1 satellites are able to maintain the objective tens to hundreds of meters apart when in experiment mode. The experiment mode will primarily be used for the scientific experiments of reflecting the starlight between the two satellites. An experiment formation between the two satellites will be selected, due to the mission requirements these will be between tens to hundreds of meters apart. This will be done so that a multitude of tests will be conducted to ensure that the satellites maintain formation. This simulation will be done using the Clohessy-Wiltshire equations coded in MATLAB. The simulation will then report the accuracy of the formation to ensure that satellites do not collide. The simulation will record whether or not the formation is able to maintain a positional accuracy of 10 centimeters. The other mode that will be simulated will be the standby mode. This mode will be simulated in a similar manner to the experiment mode utilizing the Clohessy-Wiltshire equations in python and will have similar reports. Furthermore the transfer mode will be modeled in this simulation. This simulation utilized the same equation and will be coded. The purpose of this mode will be to determine the minimal ΔV usage when transferring between standby and experiment in the relative orbit elements frame and vice versa. This will also display the trajectory of the transfer.

The total orbit simulation will take the data from the formation simulation and place it into STK. This will allow a more accurate picture as to what the two satellites will look like. This is due to STK including disturbance forces and perturbations on the spacecraft that were not included in the formation analysis. This will be used to more accurately determine the true orbital information crucial for other subsystems. These will be eclipse periods, beta angle, and true trajectory of transfers.

The decommission simulation will be used to display the time of decommissioning the satellites. This will be done using STK to show the altitude over time of the drag decommissioning and the timeframe. This will also be done to ensure that the CubeSats are decommissioned in a proper time frame of under a year after operation.

3.2.2 Formation Simulation

Due to the complexity of the STARI mission and maintaining a distance of ten to one hundred meters a simulation of the formation flying capability is required. This simulation ensured that with the hardware chosen that the mission objective can be maintained. This was done utilizing the error data from ADCS angle control and components chosen. Displaying in Figure 3.2.1 is the chosen formation for standby, experiment, the points of transfer in those formations, and the trajectory of the transfer.

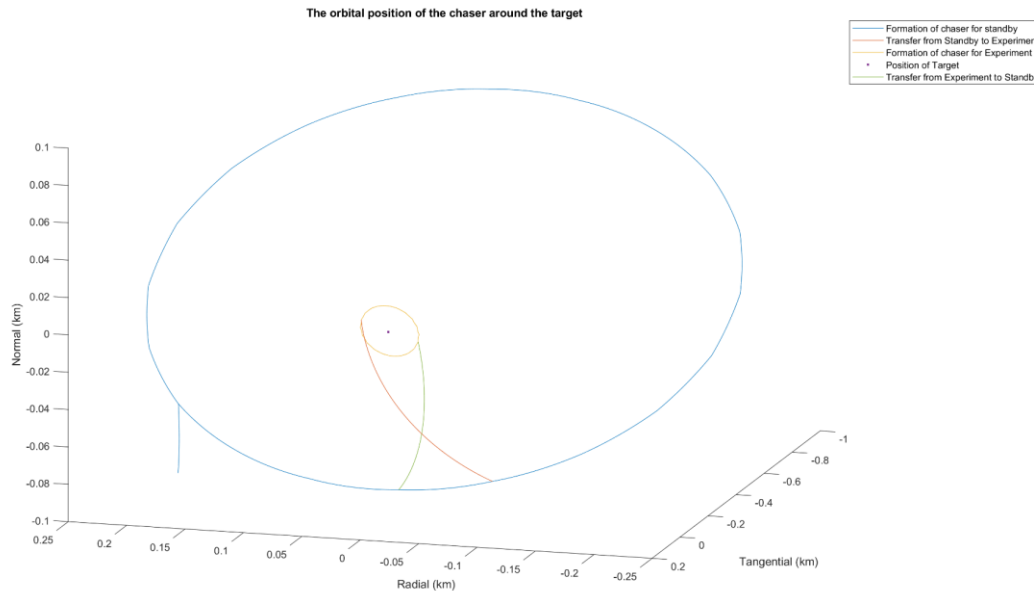


Figure 3.2.1 - Orbital Position of Chaser around the Target

As displayed in Figure 3.2.1 the formations were chosen such that the two satellites will not be more than ten to one hundred meters. The formation chosen for the standby formation was a path in the radial direction varying ± 250 meters, a circular path in the normal direction of ± 80 meters, in the tangential direction of -871.7 meters and 71.7 meters. These were chosen to maintain close communication between satellites while not being at risk of collision. The experiment formation was chosen to have a radial path of ± 25 meters, a normal direction of ± 15 meters, and a tangential direction of -36.9 meters and 16.9 meters. These values were chosen to ensure that during experiment the chaser and target maintain the mission object. Displayed is a visualization of the targeted path that the chaser would take around the target satellite in a 3D relative frame. This target formation position was used to calculate the necessary ΔV in order to maintain this position. This was done utilizing Clohessy-Wiltshire equations and the numerical solution provided [5]. Utilizing this solution yielded the ability to control the formation of the satellite by imparting the ΔV burns. From the ΔV solution the position can be determined after the burn is added. Over the course of orbit 20 burns are used to maintain the appropriate formation. Due to using the Clohessy-Wiltshire equations the expected position can be modeled over time and thus can be modeled with sensor error. This is displayed for the standby formation in Figure 3.2.2.

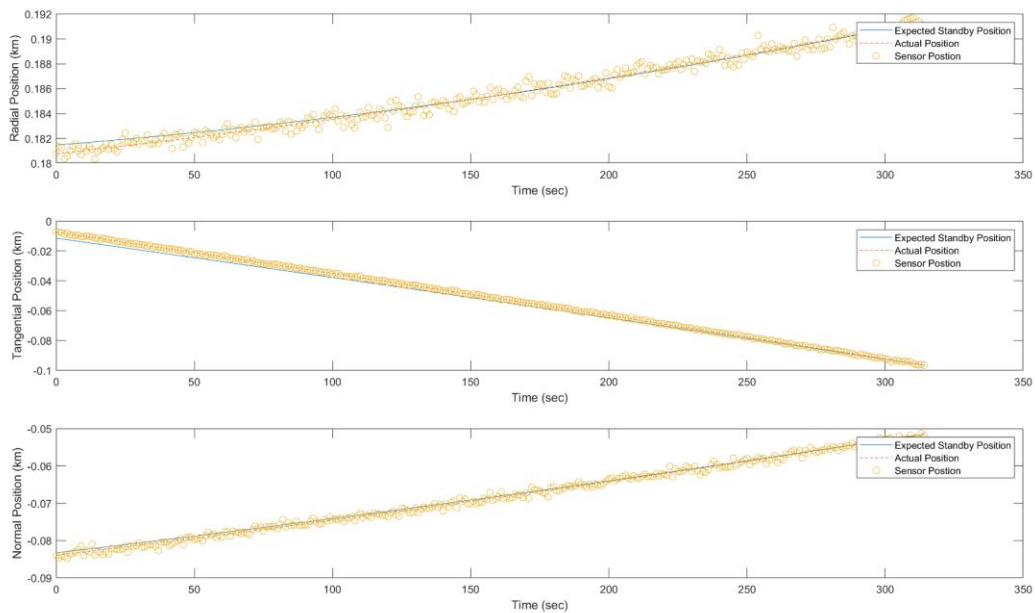


Figure 3.2.2 - Orbital Position of the Chaser in the Standby formation with sensor error

As displayed in Figure 3.2.2 is the expected radial, tangential, and normal position of the chaser along with the value the sensor is reporting and the actual position of the chaser. As can be seen the chaser is able to maintain close proximity to the expected values of the formation. The maximum position error was 9 m with an average position error of .2 m. This does not fall within the errors required for the mission.

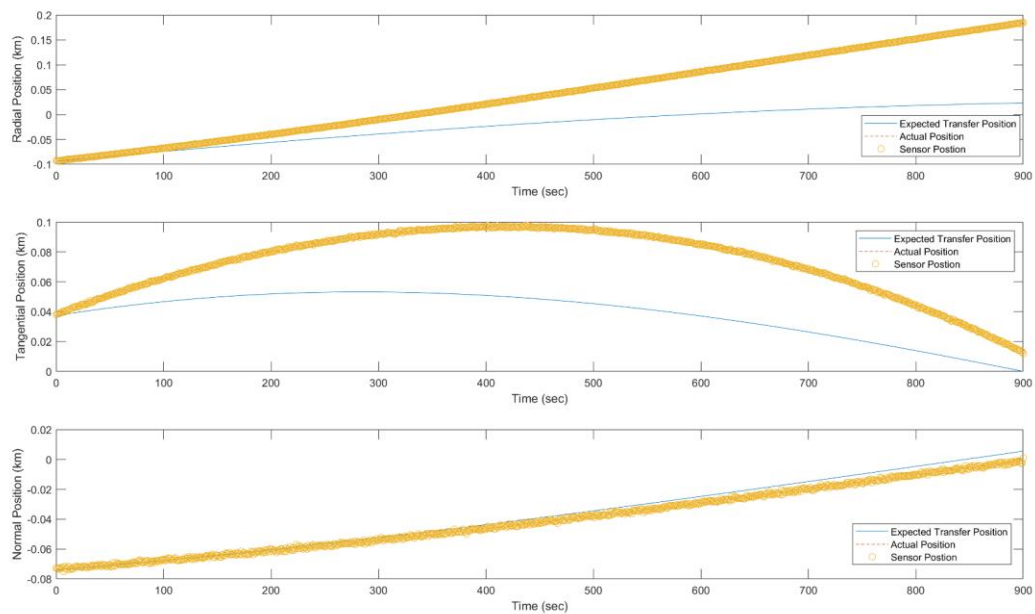


Figure 3.2.3 - Orbital Position of the Chaser in the Transfer formation with sensor error

As displayed in Figure 3.2.3 is the expected radial, tangential, and normal position of the chaser along with the value the sensor is reporting and the actual position of the chaser. Due to the transfer only using one burn there is a much larger error with the expected position. This is evident with the average position error being 59.2 meters. This does not meet the required accuracy of 10 centimeters. Due to this maneuver having only one burn over a long course of time the angle control error is more pronounced. To mitigate this error corrective burns should be employed in order to maintain proper trajectory.

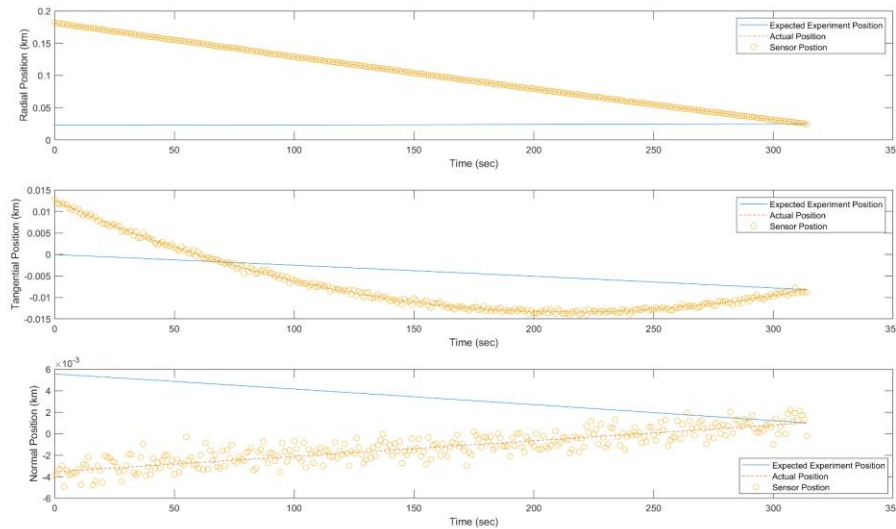


Figure 3.2.4 - Orbital Position of the Chaser in the Experiment formation with sensor error

As seen in Figure 3.2.4 the initial seconds of the experiment formation there is a larger error in the position. This is largely due to the transfer maneuver and the correction from that. As the formation is given time to settle the average position error decreases significantly. Over the course of the experiment, the average position error is 3.7 meters. This is above the mission objective however this happens due to the error on the transfer maneuver. Furthermore the minimum CubeSat distance during the experiment formation is 22.0 meters which is within the specified 10 to 100 meters of separation.

Displayed in Table 3.2.1 is the average position error, the minimum distance of the CubeSats, and the standard deviation of the error of each formation and maneuver. As can be seen from the table the transfer has the largest error with the experiment and standby having significantly less. More tuning of the transfer maneuver is required.

Table 3.2.1 - Error in formation

Formation	Average position error (km)	Minimum distance of CubeSats (km)	Standard deviation
Standby	.000202	.1084	.00012
Transfer	.0592	.1075	2.1949
Experiment	.0037	.0220	.8148

3.2.3 Total Orbit Simulation

The total orbital simulation takes the predetermined conditions of the orbit and utilizes STK to create a more defined picture. STK utilized aspects from the formation flying simulation and refined them into being more realistic. This was done because STK includes perturbations and other disturbance forces. The main aspect from the formation flying simulation that was tested was the transfer from Standby to experiment. When testing this transfer trajectory there was a 7 meter distance error from the expected position. This is above the prescribed 10 centimeter accuracy. This means that there must be a second corrective burn during the transfer arc to ensure that the chaser maintains the determined trajectory.

STK was used to determine factors that were used for other subsystems such as the eclipse period and the beta angle of the satellite. From STK the eclipse period averaged around 36 minutes of the orbit. This will be very useful information for the thermal management subsystem and power subsystem. The eclipse period is the time spent in the shadow of the earth so power generation will not be available and thermal management is paramount. During this time an experiment should not be attempted due to the power requirement. Knowledge of when this time occurs determines when transfers should be attempted. STK was used to determine the beta angle over the course of a year of operation. The beta angle is the angle between the satellites orbital plane around the earth and the geocentric position of the sun. These values are useful for thermal management and power generation. Displayed in figure 3.2.6 is the beta angle over the course of a year.

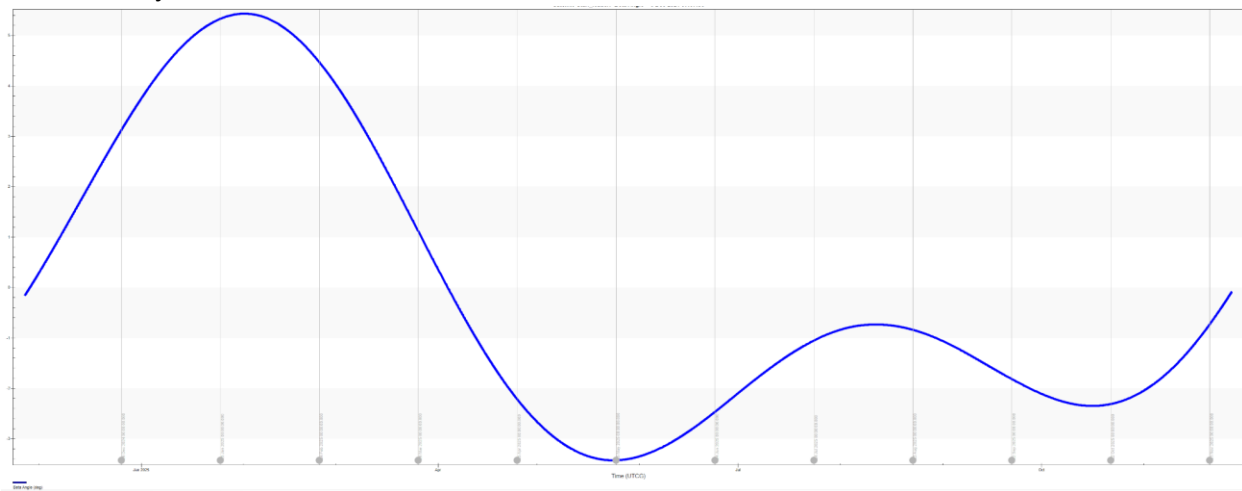


Figure 3.2.5 - STARI Beta Angle

3.2.4 Decommission Time

The decommissioning time simulation used the values given from the mass budget table along with the area provided by the drag sail to calculate the time of decommissioning of the

satellite. This was done by using STK's lifetime software. The predicted time of decommission is 146 days which is within the time frame of a year of deorbiting. Figure 3.2.6 is the height of apoapsis in yellow, height of periapsis in red, and eccentricity in blue over time when decommissioning.

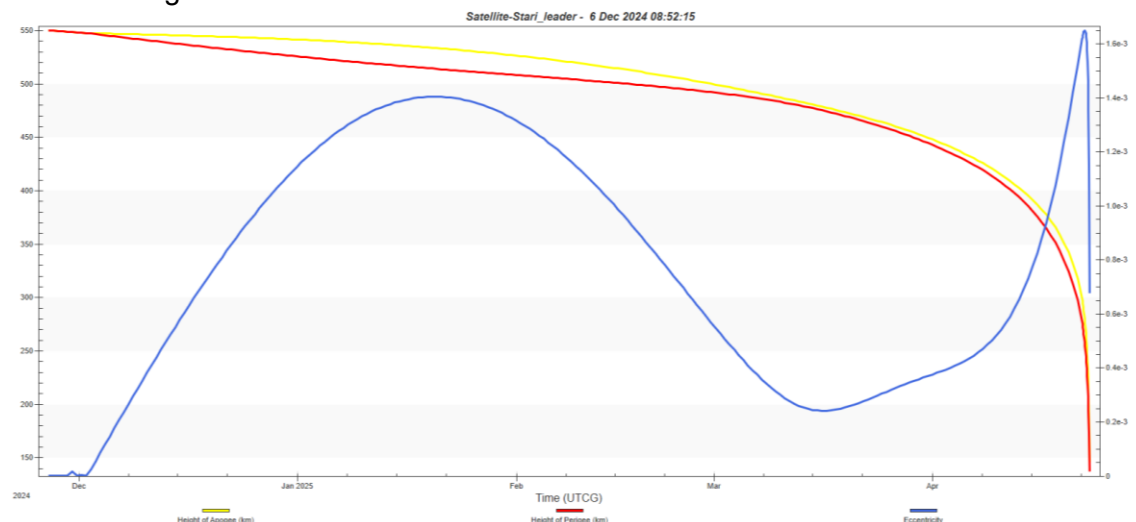


Figure 3.2.6 - Altitude of Stari when Deorbiting

3.2.5 ΔV Budget

The ΔV budget has been changed from the PDR and has been finalized to reflect the chosen formations. This has changed from the PDR as this is more specific to the formation maintenance and is more personalized to the mission. The ΔV budget utilized the average amount of ΔV used in each formation and found the total amount necessary to maintain for one standby to experiment maneuver. Displayed in Table 3.2.2 is the finalized ΔV budget for maintaining a formation.

Table 3.2.2 - ΔV Budget for Formation Maintenance

Formation	Average ΔV usage (m/s)	Total number of times used per experiment cycle	Total (m/s)
Standby	.0419	20	0.8380
Transfer	.0284	2	0.0568
Experiment	.05180	20	0.5180
Total			1.413 per experiment cycle

3.2.6 Risk Mitigation Matrix

Due to the nature of formation flying the CubeSats will be flying relatively close together during experiment. This is the main concern of the mission with respect to flight mechanics as this would prove the most catastrophic failure. This is due to the mission being a failure while also creating space debris that could endanger other satellites. This could happen for many reasons such as an error in simulations, sensor data being behind, and improper communication between the satellites.

Another prevalent risk that the satellites are exposed to are the separation from the launch vehicle failure. This could have a moderate impact on the mission due to there being an error in when deploying the CubeSats. This is more prevalent due to there being two CubeSats that need to be deployed relatively close together. Due to the error prevalent when deploying satellites extra ΔV will be allocated.

Other risks include the ΔV not being allocated correctly and solar flare. The ΔV not being allocated correctly can provide a risk due to errors in planning and not allocating enough ΔV for an emergency to take place. Solar flares can pose a risk to the on board computer systems as well as the payload. These are most prevalent during the solar maximum which occurs every 11 years. The sun recently entered its maximum period on October 15, 2024, and will continue for a year [6]. Assuming Stari doesn't launch within that year the risk of a solar flare is drastically decreased.

Table 3.2.4 - Risk Mitigation Matrix

Risk ID/Title	Severity Score	Trend	Mitigation	Verification
001 Collision of Satellites	High	→	If communication between satellites is broken ensure that there is an anticollision flight trajectory that will trigger.	Anti-collision software triggers when satellites lose contact in experiment formation. It will use the last known location of the target satellite to know where to apply thrust.
002 ΔV is not allocated properly	Moderate	→	Run multiple simulations with a high degree of accuracy and precision, that will also anti collision maneuvers and errors in launch vehicles. Ensure that there is extra ΔV allocated for emergencies.	Simulations detailing non ideal scenarios to ensure that there is a proper excess of ΔV in case of emergencies.
003 Separation from launch vehicle error	Moderate	↓	The Falcon 9 comes with error in the launch vehicle deployment of around 10 kilometers for perigee and 15 kilometers for apogee. This will be accounted for and managed.	Extra ΔV allocated due to the error from the launch vehicle will be tested to ensure that it will be enough to maintain mission objectives.
004 Solar flare	Moderate	→	The solar maximum happens roughly every 11 years which is when solar flares are the most common for LEO satellites. The last solar maximum was October 24 2024. Due to this it is unlikely to occur.	Accepted as is

3.3 Attitude Determination and Control System (V. Bindal)

3.3.1 Overview

The ADCS system controls and stabilizes the spacecraft's orientation during the mission. According to the mission details, advanced simulations and calibrations are necessary to meet mission accuracy requirements. The calibration simulation aims to orient the sensors such as star trackers and sun sensors while also ensuring the data from the GPS is accurate. The GPS and IMU sensors may need to be calibrated multiple times throughout the mission to ensure their accuracy. Once calibrated, the CubeSat will simulate to de-tumble and avoid scenarios such as "Lost in Space". From there boundaries and conditions can be made to start creating an open-loop 3D simulation. This simulation must account for noise and ensure the sensors meet the pointing accuracy for formation flying. A control algorithm must be chosen and implemented such as the Extended Kalman Filter which helps reach the accuracy needed in a desirable amount of time. Formation flying is a requirement for this mission so simulation of both CubeSats performing the fiber injection. Ensure stable injection into a single-mode fiber continuously for at least 10 seconds and ensure CubeSats maintain an angular accuracy of 2 degrees over a distance of 100 meters. Perform the simulations multiple times with different environmental disturbances to ensure proper selection of control algorithm and mission reliability.

3.3.2 Sensor Simulation

To begin simulation of the sensors, data is needed from specification sheets listing out errors inherent in the sensors. Gyroscopes and accelerometers tend to have random walks such as bias instabilities that build up over time leading to a greater error if not calibrated frequently. [8][9][10][11] Shown in Table 3.3.1 are all sensors in use. The error of the sensor of the Pulse-40 was calculated using the sensor range multiplied by the scale factor error and non-linearity which are then summed up. The random walk was given but was converted from degrees per $\sqrt{\text{hour}}$ to rad per $\sqrt{\text{second}}$. Update rates were also accounted for to ensure time steps within the simulation were accurate. When creating the simulation, it is important to convert 3-sigma values to a 1-sigma value and then find the square root of the value to get a variance.

One thing to remember is that the star tracker points in a specific direction based on the spacecraft's approximate attitude provided by coarse sensors like gyroscopes. Hence if the star tracker is not pointed towards a star or the field of view is obstructed it can not give accurate angle measurements. From here an open-loop simulation can be made. MATLAB software will be used to create the simulation as it has great capabilities in implementing algorithm and complex matrix calculations. The script simulates the open-loop dynamics of a spacecraft using sensor data in a 3D environment. The script achieves this by modeling true dynamic quantities such as acceleration, velocity, position, orientation angles, and angular velocity while introducing realistic noise and biases to simulate sensor readings.

Table 3.3.1 - Sensor Error Data

Sensor	What does it measure?	Error of sensor (Variance)	Random Walk	Update Rate
ST200 (Star Tracker)	Attitude Determination (Angle)	10 arcseconds	N/A	5 Hz
Pulse-40 (IMU Sensor)	Gyroscope (Angle Rate) Accelerometer (Linear Acceleration)	Gyroscope - (0.1154 rad/s) Accelerometer - (0.3705 m/s ²)	Gyroscope - (2.32*10 ⁻⁵ rad/√s) Accelerometer - (3.33*10 ⁻⁴ m/s/√s)	100Hz
AsteRx-m3 Pro+ (GNSS system)	Position Velocity	Position (x and y = 0.632 meters, z = 0.837 meters) Velocity = 0.3 m/s	No need since the position is updated using satellites	10 to 100 Hz

To create an accurate model, simple equations of motions are used. First, acceleration is modeled as random noise with a Gaussian distribution to simulate real-world variations in forces. The script uses sensor values from Table 3.3.1 and incorporates the equations of motions listed below. A time step is related to the update rate and the sensor it is related to. For example, the time step for position is 10 hertz which is once every 0.1 seconds but acceleration is found every 0.01 seconds.

To accurately model proper attitude control the following Euler angles (roll, pitch, yaw) are used. Euler angles provide a structured way to represent the spacecraft's orientation in 3D space, offering intuitive insight into its rotational states. Euler angles are derived using quaternions. [12] Quaternions are used to avoid the singularities such as gimbal locking (check Appendix C) and computational inefficiencies of Euler angles. Each quaternion (q) is defined as a 4-element vector $[q_0, q_1, q_2, q_3]$, where q_0 is the scalar part and q_1, q_2, q_3 is the vector part. The simulation computes quaternions iteratively by integrating angular velocities (ω) using quaternion multiplication. At each step, the quaternion is updated to reflect the spacecraft's new orientation, which is then converted to Euler angles. The calculated Euler angles represent the spacecraft's orientation in a readable form suitable for visualization and analysis. These angles are further augmented with random noise to simulate real-world sensor imperfections, allowing the evaluation of sensor accuracy and the system's ability to handle noisy inputs.

To compute how the spacecraft's orientation changes over time, the angular velocity vector relates to the time derivatives of the Euler angles, and T_{euler} provides this relationship. The transformation matrix becomes singular when the pitch angle becomes $\theta = \pm\pi/2$. This is known as gimbal lock, where the rotation axes become aligned, causing a loss of one degree of freedom. [13] Quaternions avoid gimbal lock by representing rotations in a four-dimensional space,

bypassing the sequential axis rotations of Euler angles that can lead to axis alignment and loss of degrees of freedom. Additionally, quaternions provide smooth and continuous rotation representation, computational efficiency, and numerical stability, making them essential for precise orientation management. As seen in Figure 3.3.1 is the output of the sensor angle vs the true angle. Here alpha is set to: $\alpha = [0.5, 0.3, 0.1]$

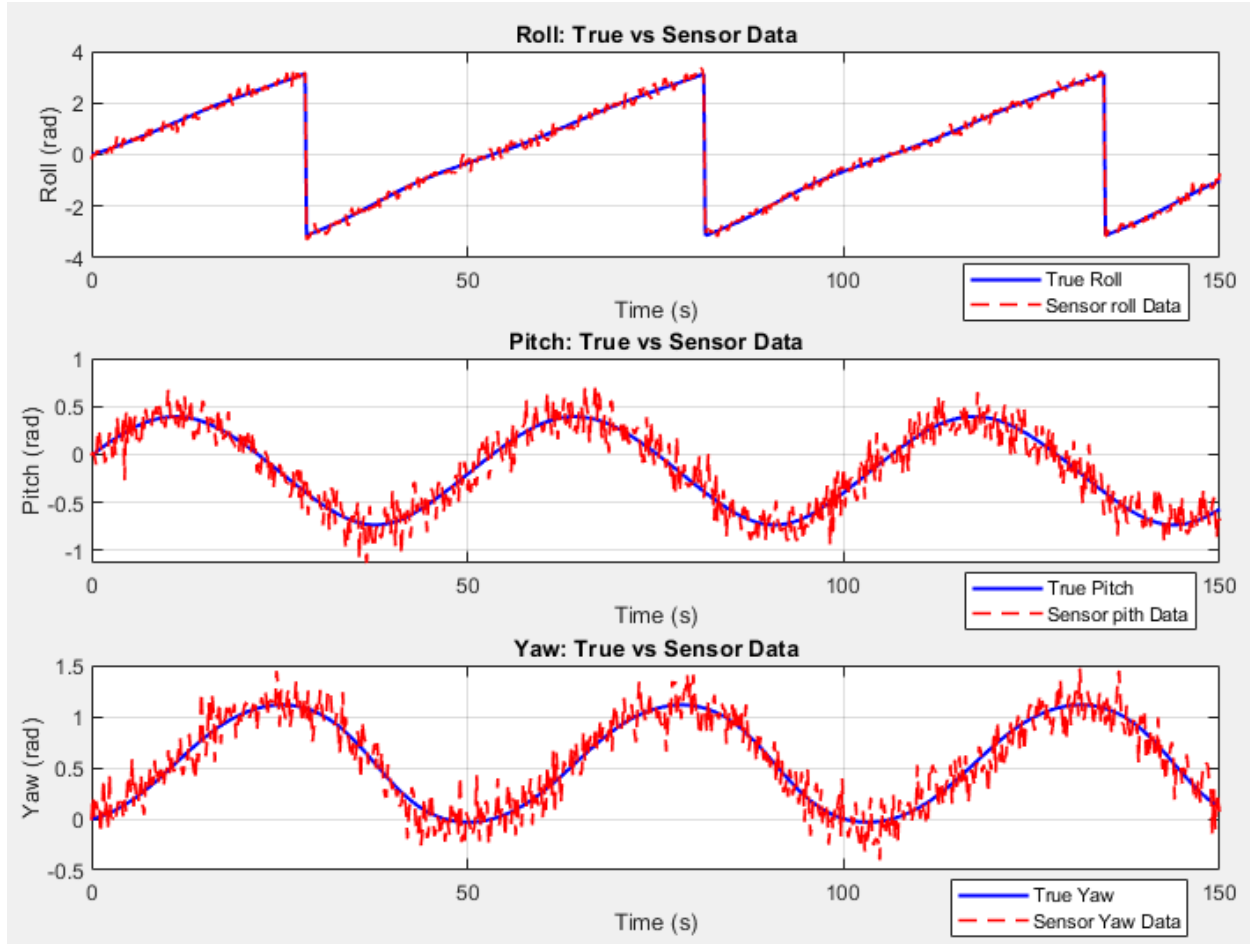


Figure 3.3.1 - Sensor angle vs. True angle

It is easy to see that the sensor data is more accurate with linear changes especially looking at roll. The model also makes realistic sense since such a small acceleration with no initial velocity means that it should take time to complete one rotation. Since the simulation only models 150 seconds, acceleration and gyroscope drift are not given enough time to properly be shown in the plot. Figure 3.3.2 shows the drift and its accumulation over time.

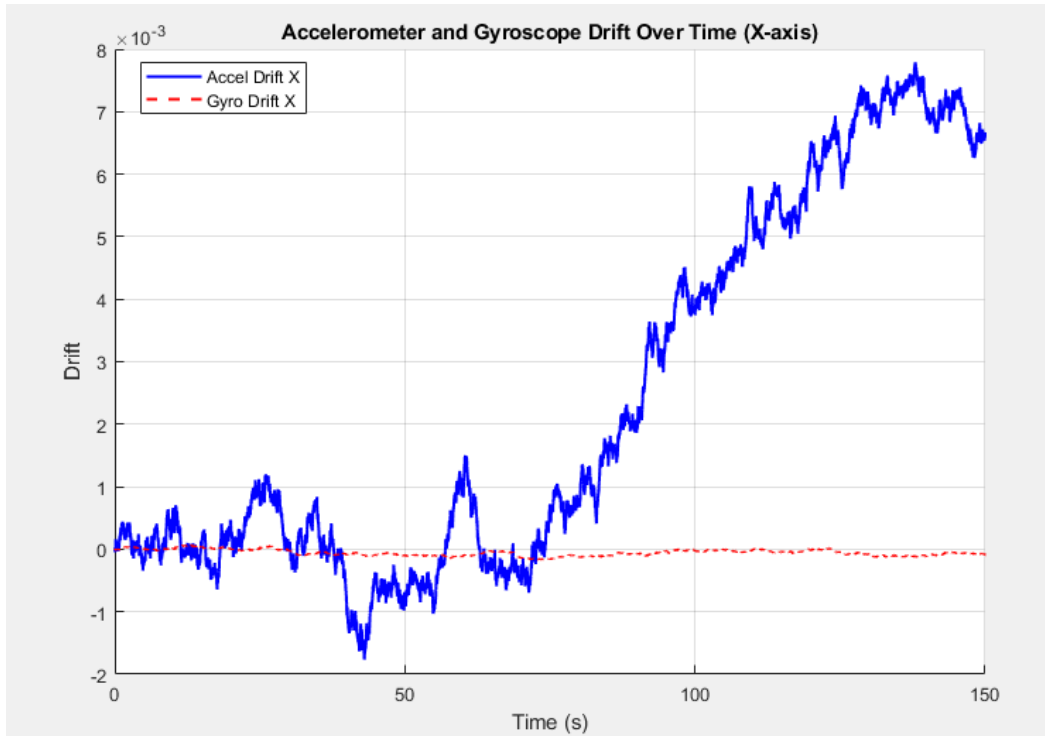


Figure 3.3.2 - Acceleration and Gyroscope drift

Due to this increasing rate of error, the IMU sensor will need to be calibrated frequently to ensure accurate orientational data. Some of this error will be filtered out by the implementation of the Extended Kalman Filter as it would combine sensor data with other reference inputs such as the GNSS to continuously refine bias estimates. Using external references to update the bias is crucial, and can be done with star trackers and GNSS. From here a closed-loop simulation can be created and implement a control algorithm.

3.3.3 Closed-Loop Dynamics Simulation

[14] The Extended Kalman Filter (EKF) implemented in the script estimates the spacecraft's dynamic state, including position, velocity, acceleration, orientation angles, and angular velocities, using noisy sensor measurements. Unlike a standard Kalman Filter, which assumes linear dynamics, the EKF handles nonlinear systems by linearizing them at each time step using a first-order Taylor expansion. The EKF operates in two main phases: the prediction phase, which projects the current state forward in time based on a dynamic model, and the update phase, which refines this prediction using incoming sensor data. In the prediction phase, the state transition model describes how the state evolves over time. This is represented by a state transition matrix, F , which incorporates dependencies between position, velocity, and acceleration, as well as orientation and angular velocities. Position is updated using the previous velocity and acceleration over the time step, while orientation is updated based on angular velocities.

The MATLAB script used to demonstrate the EKF uses extracted data from the open loop simulation to account for drift and sensor error. In Figure 3.3.3 position is graphed to easily see the EKF at work. The true position data is a product of random acceleration generated. It is clear

to see that EKF does give a better estimate compared to the raw sensor data but could use some refining when it comes to the process noise (adjusting values within the Q matrix).

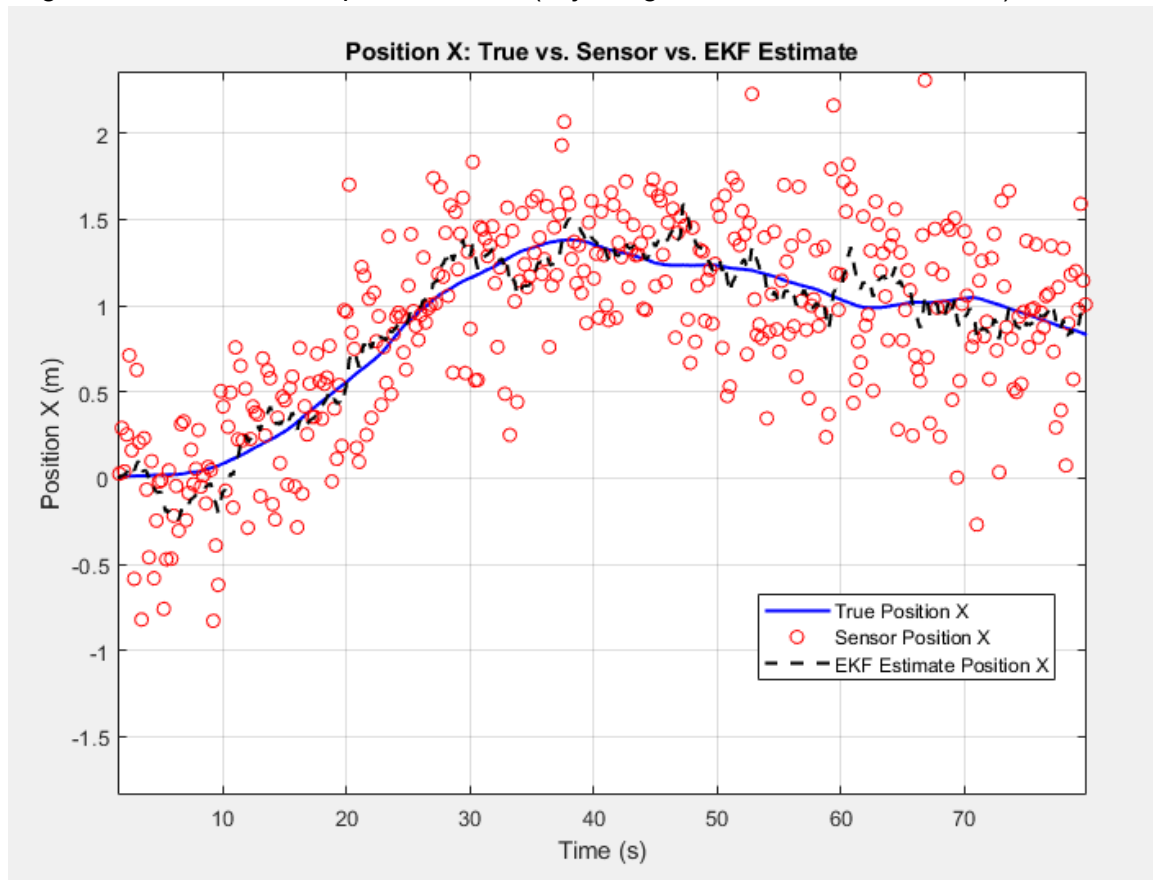


Figure 3.3.3 - EKF position data

Similarly, angle is also filtered through the EKF but since the star tracker has much better accuracy than the GPS, the graph in Figure 3.3.4 shows an exaggerated sensor error. The star tracker has an accuracy of 30 arcseconds but to ensure that the EKF works properly the input error was given at 3000 arcseconds. This is with the star tracker being properly calibrated and pointing towards a star.

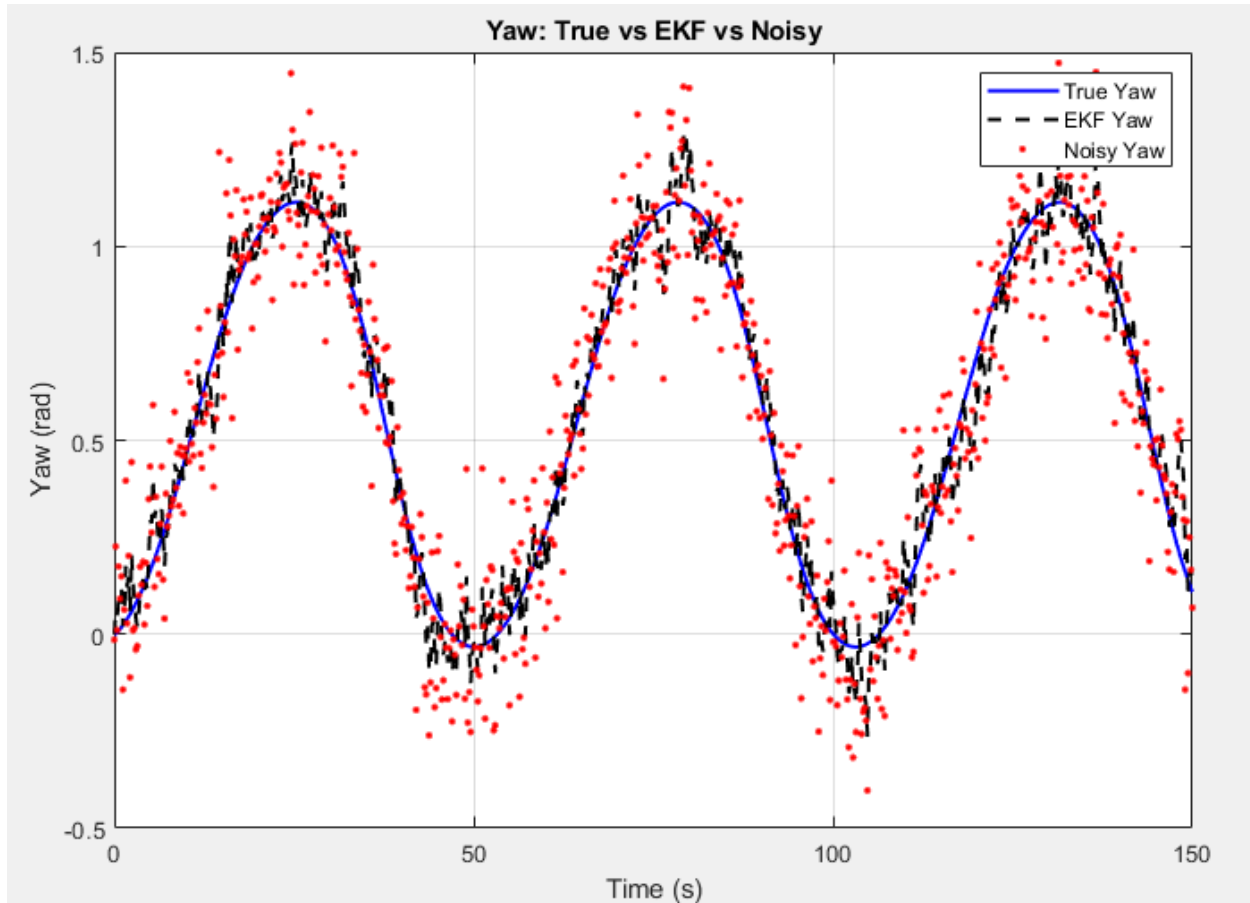


Figure 3.3.4 - EKF Yaw data

With the EKF working properly, the simulation can be repeated multiple times and find the standard deviation of the error from the true data. This gives Table 3.3.2 which contains the standard deviations of error from the sensor and then from the EKF. Sensor error should match almost perfectly with the input error. These errors are found after averaging 10 simulations.

Table 3.3.2 - EKF vs Sensor Error

Kinematic Quantities	Sensor Error (σ)	EKF Error (σ)	Increase in accuracy (%)
Position X and Y (m)	0.6267	0.2120	66.2%
Position Z (m)	0.8653	0.3176	63.3%
Velocity (m/s)	0.2981	0.1266	57.5%
Acceleration (m/s ²)	0.3065	0.1390	54.6%
Angle (rad)	1.406e ⁻⁴	1.142e ⁻⁴	19.8%
Angular Velocity (rad/s)	0.1179	0.0855	27.5%

The Extended Kalman Filter definitely increases the accuracy of the sensors and handles the gyroscope and accelerometer drift well. Accuracy can be improved by making small changes to the process noise matrix and the measurement noise matrix within the EKF. Fine-tuning these matrices ensures the filter strikes a better balance between trusting the sensor measurements and relying on the prediction model. The EKF demonstrates significant accuracy gains across all measured quantities, with the most notable improvements in position and velocity. Even then, the current simulation does not meet the 10 centimeter level position accuracy needed for the mission. This can be mitigated in many ways. Extra star trackers to reduce the initial sensor error or more advanced GPS systems essential ones using differential GPS techniques. It would also be possible to implement an Unscented Kalman Filter that dynamically adjusts process and measurement noise matrices based on real-time data. Another

3.3.4 Angle Control Simulation

From the data sheet of the RW400, the control accuracy of 1% of the rpm gives the basis to model the pointing accuracy of the CubeSats. With a maximum of 5000 rpm and known moments of inertia from Table 3.1.3, induced velocities can be found. This helps simplify the simulation as reaction wheels can now be visualized as velocities. In Table 3.3.7 all negative values indicate that the spacecraft moves in the opposite direction of the wheel's movement. The induced angular velocity is also the max angular velocity before a reaction wheel saturates

Table 3.3.3 - Reaction wheels induced velocity

Axis	Induced Angular Velocity ($\Delta\omega$ in rad/s)	Angle Control error (rad/s)
Roll	-0.3539	± 0.00359
Pitch	-0.2818	± 0.00282
Yaw	-0.6929	± 0.00693

Table 3.3.3 takes into consideration magnetorquers. Since magnetorquers need an electrical field to operate, an average was taken when considering Earth's magnetic field in LEO (4.5×10^{-5} Tesla). Magnetorquers are crucial to angle control as they allow reaction wheels to desaturate without affecting the orientation of the spacecraft.

Table 3.3.4 - Magnetorquers induced velocity

Axis	Induced Angular Velocity ($\Delta\omega$ in rad/s)	Angle Control error (rad/s)
Roll	0.0042688	$\pm 4.2688 \times 10^{-5}$
Pitch	0.0033348	$\pm 3.3348 \times 10^{-5}$
Yaw	0.0082093	$\pm 8.2093 \times 10^{-5}$

With such low torques, magnetorquers can be used to make very precise movements but come with the drawback of taking a long time to perform maneuvers. Using reaction wheels and magnetorquers together will help cancel out each other's imperfections, creating a fast but reliable attitude control system. Though the reaction wheel and magnetorquers themselves have the ability to maintain the 2 degrees over a 100 meters for at least 10 seconds, the added error from the EKF needs to be accounted for.

[11] A MATLAB simulation was made with a chaser CubeSat and target CubeSat distanced 100 meters away. Using the EKF data previously created and implementing PID, Figure 3.3.9 was produced. Keep in mind that the angular accuracy of 2 degrees at 100 meters translates to a position accuracy of about 3.5 meters. This means that any value below the 3.5 meter inaccuracy satisfies the requirement. The lateral offset also takes into account the position error from the EKF.

Table 3.3.5 - Angle Accuracy Verification

Axis	Max Error (degrees)	Lateral offset at 100 meters (m)
Roll	0.6400	± 1.3271

Pitch	0.5667	± 1.1991
Yaw	1.2883	± 2.5659

From there PID was used to simulate one of the Euler angles, Yaw, to ensure that the CubeSat could reach its desired value in time. PID control is a widely used feedback control technique that enables systems to achieve desired behaviors like precise positioning, stabilization, or tracking a target. In the context of this mission, PID can be used to manage attitude control by adjusting angular velocity to achieve accurate pointing. The proportional term provides a control output that is directly proportional to the current error while the integral term accumulates past errors over time and generates a control signal to address long-term discrepancies between the desired and measured values. The derivative term responds to the rate of change of the error, effectively predicting future errors and applying a damping force. These terms are carefully balanced to reduce steady-state error and overshoot. With a low steady-state error, the payload has the best chance of success to complete the mission. With no overshoot, extra power is not being used to recorrect overused reaction wheels. This simulation uses a small angular velocity to realistically model the CubeSat in standby. From Figure 3.3.10 it is clear to see that the CubeSat reaches a steady-state within about 10 seconds. This can be further improved with more robust K_D, K_I, K_P values.

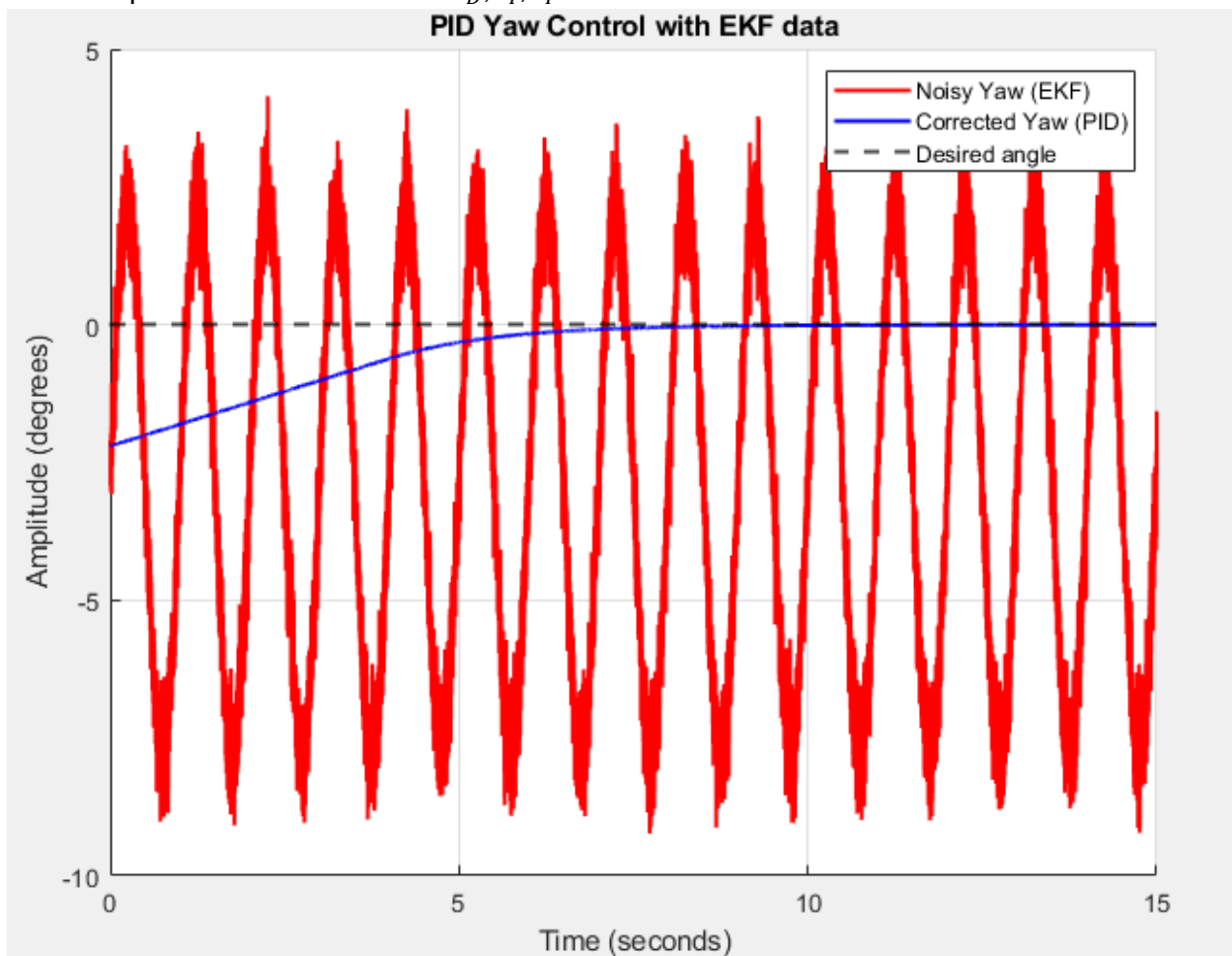


Figure 3.3.5 - PID angle control

3.3.5 Risk Mitigation Matrix

Below are the primary risks associated with the ADCS subsystem. These risks were selected based on their potential to significantly impact the mission's objectives. For instance, sensor malfunction is a critical risk, as it directly affects the spacecraft's ability to maintain the 2-degree angular accuracy tolerance necessary for the mission's success. Additionally, inaccurate attitude control for formation flying poses a challenge, as it could compromise the payload's ability to transfer starlight. Power limitations for control system components, such as reaction wheels and magnetorquers, add another layer of complexity. If insufficient power is allocated, it could lead the CubeSats lost in space or make it impossible to get out of a tumble.

Environmental factors also present considerable risks to the mission. External disturbances, such as gravity gradient torque, solar radiation pressure, or atmospheric drag, can introduce unanticipated forces on the spacecraft. Though the disturbances may be small, over time spacecraft will be forced to perform correction burns. With calibration drift being another concern, it could lead to cumulative errors in attitude determination over time, which would degrade the chance of a successful payload performance. These risks emphasize the critical need for careful design, extensive testing, and validation of all subsystems to ensure that the spacecraft can reliably meet its operational objectives and deliver the scientific and technological outcomes expected of the STARI mission.

Table 3.3.6 - EKF angle data

Risk ID/Title	Severity Score	Trend	Mitigation	Verification
001 Oversaturated Reaction Wheels	High	→	Implement advanced momentum management techniques, such as desaturation strategies using magnetorquers. Regularly monitor wheel speeds to prevent reaching saturation.	Conduct simulations of reaction wheel operations under nominal and disturbed conditions. Validate desaturation algorithms in hardware-in-the-loop tests.
003 Unaccounted External Disturbances	Moderate	→	Account for drag, radiation pressure, and gravity gradient torque in control algorithms, and use disturbances as part of the control model.	Validate robustness in simulations with environmental disturbance models and onboard experiments.
004 Calibration Drift	Moderate	↑	Implement periodic recalibration protocols and use machine learning to detect and correct drift trends.	Conduct on-orbit calibration after major maneuvers or significant operational phases.
005 Unaccounted Noise in Sensor Data	Low	→	Use filtering techniques like Kalman filters to mitigate the impact of noise and ensure accurate attitude estimation	Validate sensor performance with simulated noisy data and test ADCS algorithms in hardware-in-the-loop.

3.4 Commands & Data (A. Cohen)

3.4.1 Commands & Data Analysis Overview

The commands and data subsystem is responsible for the electronic architecture which coordinates and controls the satellite. The system stores data to be transmitted to the receiver on the ground and autonomously controls the craft based on onboard sensors and received commands. The satellite must be able to determine and adjust its attitude and direct power where necessary without commands from the ground. In order to ensure the systems will function within parameters, analytical and dynamic simulations will be run. The connections between electronic systems will be mapped and added below.

Several systems such as ADCS, thermal management, power, and the payload will produce data that will be stored on board until it can be transmitted to the ground station for analysis. This can be simulated by summing the data produced over a period of time as the craft goes through various modes where the amount of data changes. In the event the data storage on the satellite has reached capacity and it is unable to contact the ground, some data must be deleted. Determining what data is lost is based on its source, engineering data likely being less critical to the mission than the data from the payload.

A state machine diagram is added to show how the satellite will respond to input to function autonomously. The craft must be able to calibrate internal systems and coordinate with the other satellite to stay in formation without control or input from the ground.

3.4.2 Connections System Diagram

In order to control the systems on board the satellite, the computer must be connected to each system to communicate data and commands. For systems onboard the satellite connections will be made using wires and pin connections.

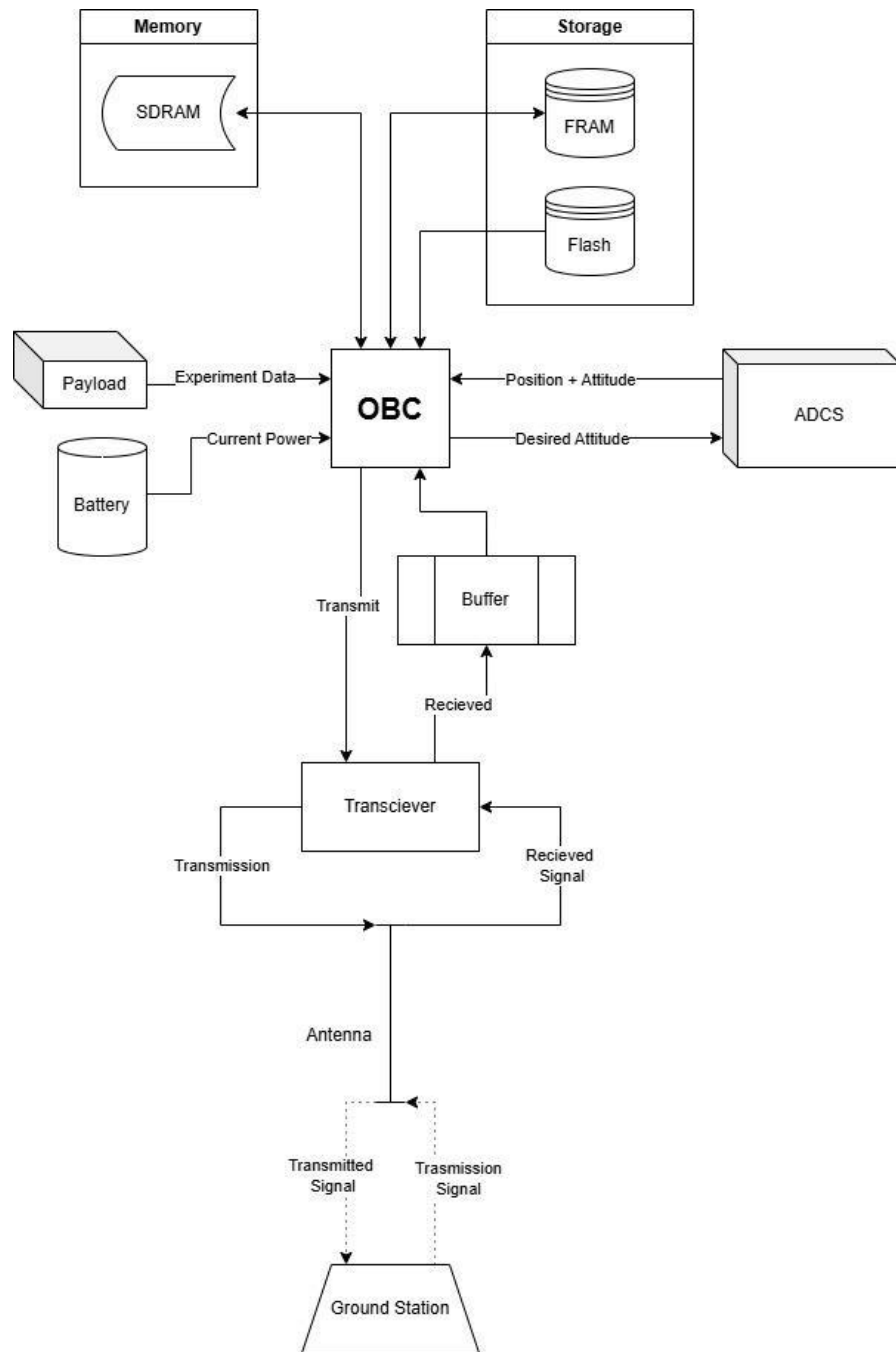


Figure 3.4.1 - Connections Diagram

The figure shows the data connections between the computer and other systems. The FRAM acts as the memory, storing data for short periods of time. Long term storage stores the data that will be transmitted in the SDRAM. The flash storage contains instructions and programs and is read-only, meaning the stored information will not be edited or deleted and no data is added. The power system measures the current amount of power stored in the battery. The ADCS system relays the current position and attitude of the satellite and the OBC provides

the system with the desired attitude so the reaction wheels can adjust the craft to the correct orientation.

3.4.3 Modes

In order to autonomously function the satellite must respond to sensor input and switch to the appropriate mode. A mode is the functionality of the satellite, with each mode determining how the satellite will operate and function. A table of different modes and the functions they perform along with the criteria required for the satellite to switch to the respective mode is shown below.

Table 3.4.1 - Modes

Mode	Function	Criteria
Standby	Monitors Sensors Checks for errors or corruption Maintains standby orbit	Default mode
Communication	Directs antenna toward ground station Transmits data Receives commands	In Communication Window
Transfer	Finds position of other satellite Moves satellite into position within parameters	CubeSats can successfully communicate
Experiment	Directs satellite towards star Enter experiment orbit Uses ADCS to keep payloads pointed at each other	Experiment command received

Once disconnected from the launch vehicle, the satellite will start in initialization mode (Figure 3.4.3) before entering standby mode. A state machine diagram of the operation of the satellite is shown below (Figure 3.4.2).

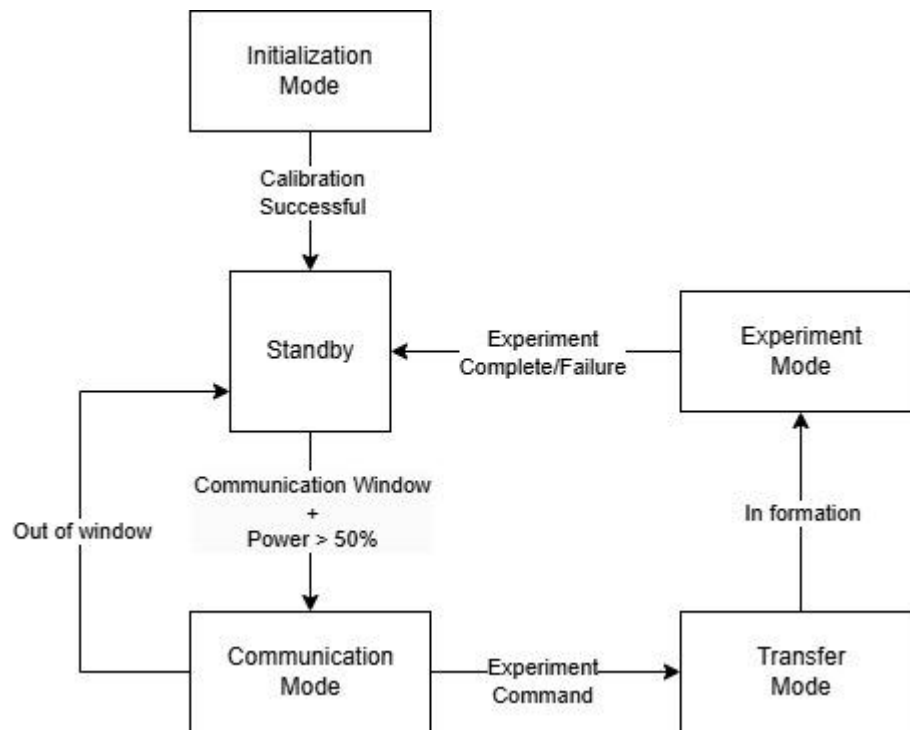


Figure 3.4.2 - Overall State Diagram

Standby mode monitors the onboard sensors, receives radio transmissions, directs solar panels to the sun, and constantly checks the position of the satellite in orbit. The position of the satellite in orbit is determined based on a Kalman analysis of the position given by the ADCS system.

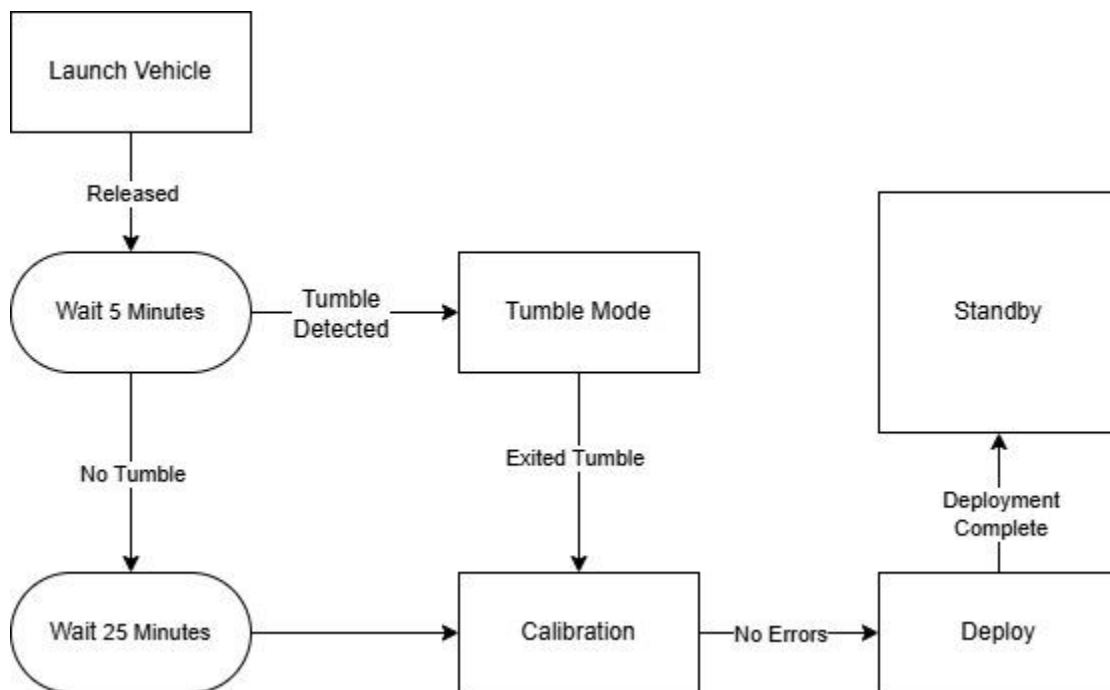


Figure 3.4.3 - Initialization Diagram

The CubeSat must wait until it is a safe distance from the launch vehicle before it checks onboard sensors for tumble. If tumble is detected the reaction wheels are used to attempt to exit tumble. If no tumble is detected the satellite waits to gain additional distance from the launch vehicle then begins calibration of the onboard sensors.

If the position of the satellite is within the communication window and the power stored in the battery is greater than half of its total capacity, the OBC switches to communication mode (Figure 3.4.4). Communication mode will direct the antenna toward the ground and power on the transmitter. Charging is placed before the communication mode so if the satellite is in sunlight the panels are directed towards the sun before the attitude is changed so the panels remain in as much light as possible, assuming the attitude will be adjusted along the shortest angle.

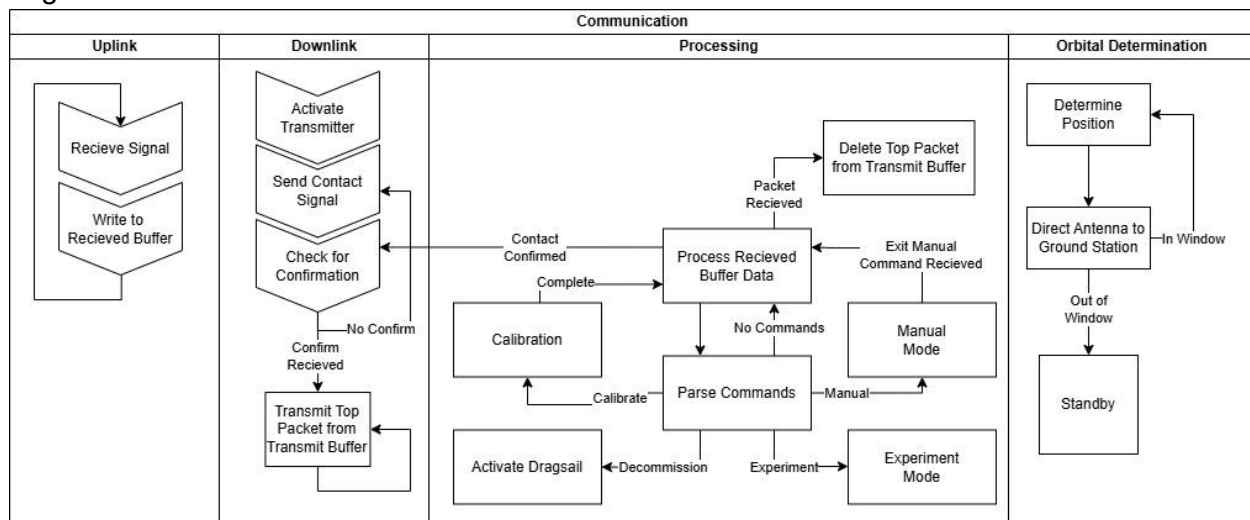


Figure 3.4.4 - Communication Mode

The downlink activates the transmitter and sends a signal to the ground station to confirm that communication is possible. If the satellite does not receive a signal from the ground station after a short period, another attempt to contact the ground station is made. This repeats until a signal confirming contact is received or the satellite exits the communication window. Once a confirmation from the ground has been received the first packet of data from the queue is transmitted. The satellite waits for a signal confirming the packet has been received and the data can be deleted from the OBC storage. This is repeated with the next packet on the queue, now the first packet. The queue is ordered based on importance to the mission; mission-critical information such as data produced by the experiment is given higher priority than engineering data, and will be placed higher in the queue and sent first.

The uplink, run in parallel with the downlink, monitors messages from the ground station, and sends the signal to the buffer for the OBC to process into commands. Processing reads the data from the receiving buffer and parses the output for commands. A command to start manual mode forces the satellite to act only based on received commands until a command is given to exit the mode. When the satellite has reached the end of its mission the decommission command activates the dragsail, causing the satellite to deorbit and burn up in the atmosphere.

3.4.4 Inter-Satellite Coordination

The CubeSats must be in contact to function together and perform the experiment needed for this mission. The satellites will be in contact through radio transmissions of commands from a commanding satellite and data from a following satellite. The commander will transmit necessary commands to the follower when commanded by the ground station or when internal systems require coordination. During the communication mode the commander CubeSat will transmit to the ground and receive any commands. If data cannot be entirely stored on the commander, the follower will store the excess and transmit it to the commander as the commander data is deleted.

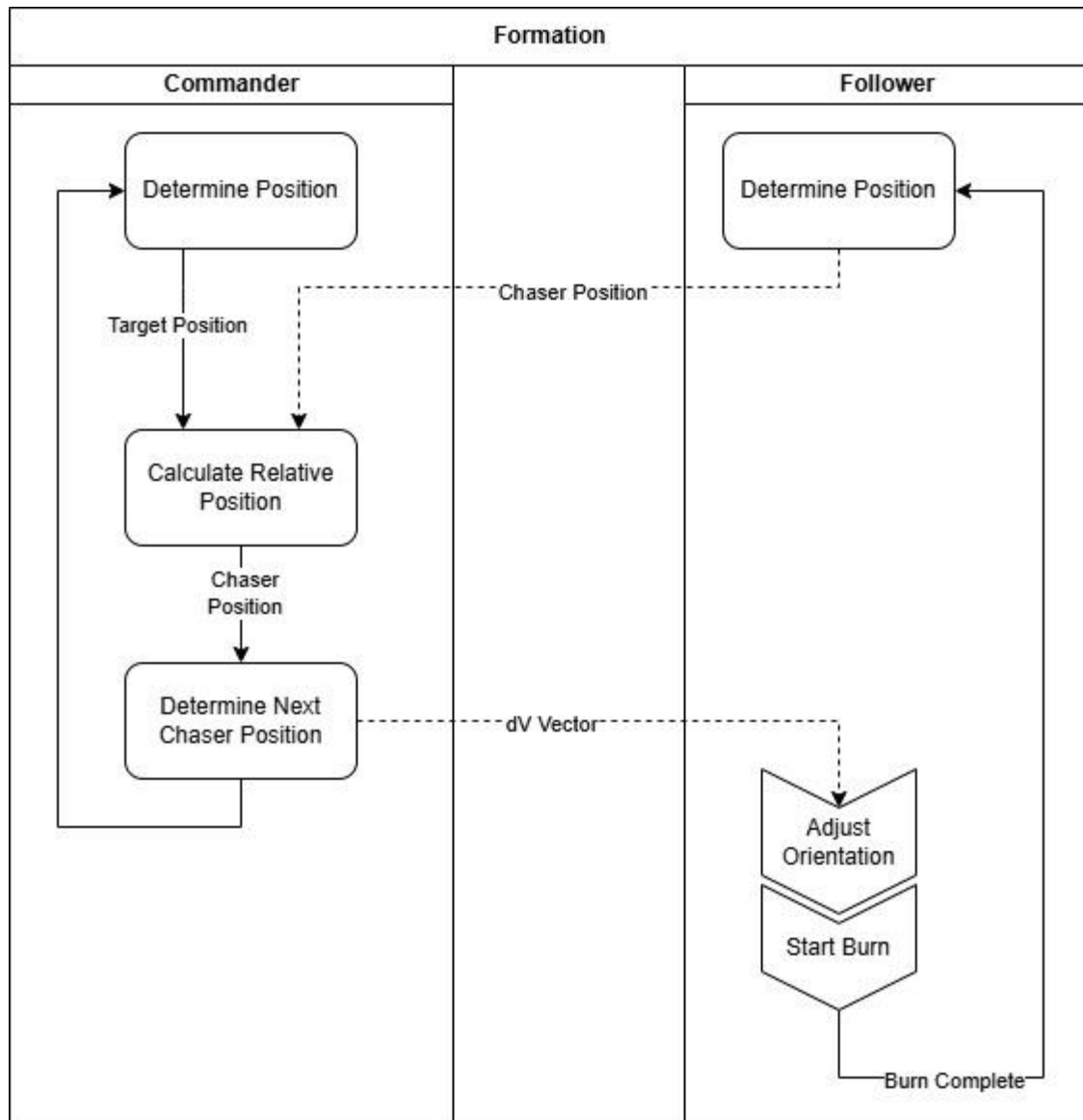


Figure 3.4.5 - Standby Formation

The commander determines the relative positions of each CubeSat then sends a command to direct the follower to the next point in the standby orbit. When the satellite receives the command to begin the experiment, the commander will send a command to the follower to change its orbit to get into a suitable position for the experiment.

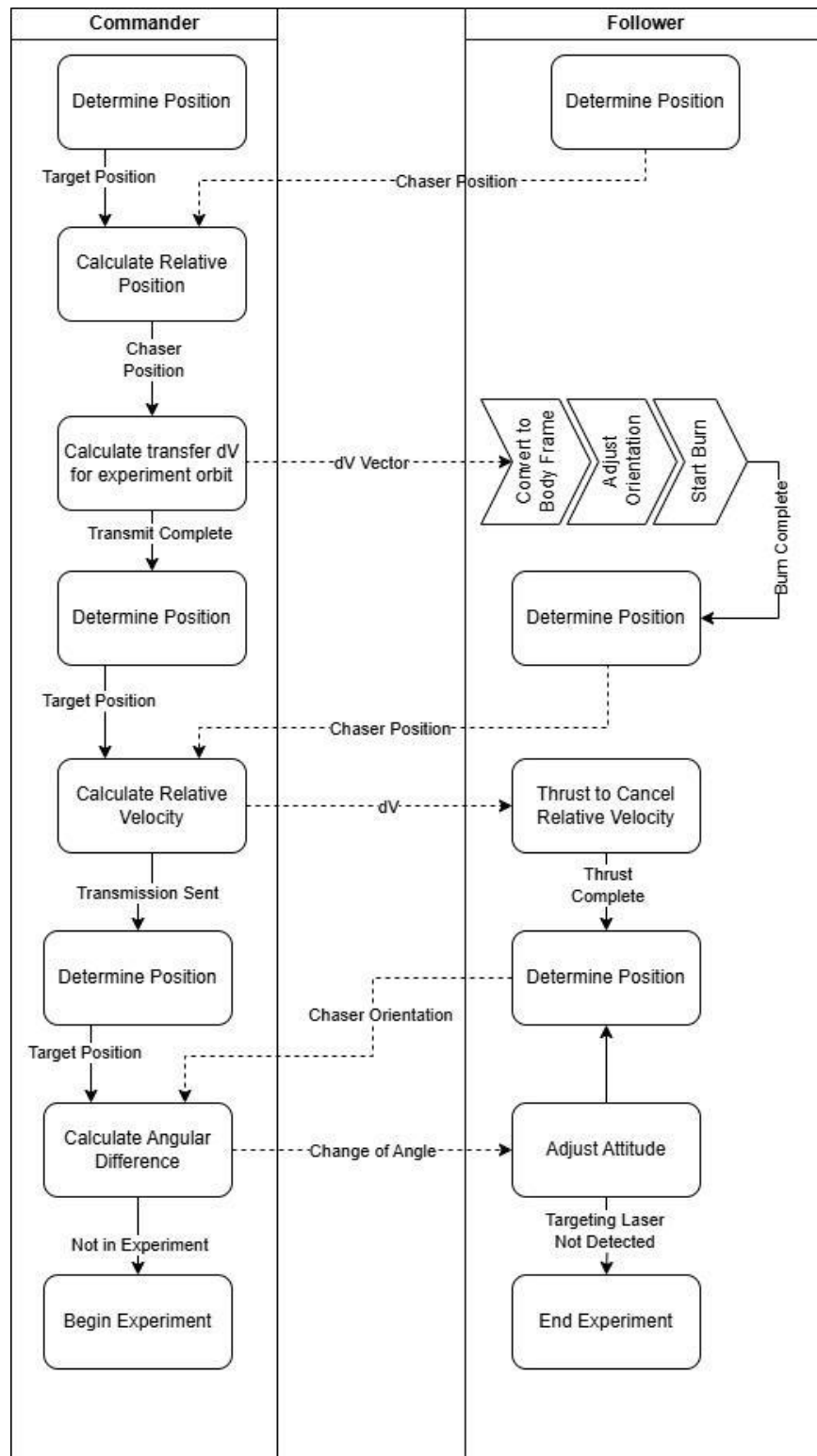


Figure 3.4.6 - Experiment Mode

From the new orbit a command is sent to the follower to match the velocity of the commander. Once the attitudes of the crafts are aligned, the experiment can begin. The follower directs starlight to the commander and the commander calculates any needed attitude

adjustment. When the commander no longer detects the targeting laser the experiment is ended and the satellites return to standby mode.

3.4.5 Data Flow Schedule

The OBC uses a Flash storage drive to hold instructions for operation, and a FRAM to hold data before it is transmitted to the ground station. The data will be collected from the onboard sensors and transmitted to the ground station for analysis. Once the data is transmitted to the ground and the ground has confirmed it has received the packet, it can be deleted, freeing the space for new data. In the event the storage is full, the lowest packet on the queue is deleted. The ADCS system produces an average of 5 B/s which is saved to the OBC to be transmitted to the ground so the path of the orbit can be tracked. The temperature sensors produce roughly 15 B/s and the OBC log is estimated to write at 50 B/s as an average over the orbit. During the experiment, the payload will produce 10 kB/s for roughly 10 seconds. During the communication window the satellite will be able to communicate around 65 GB on average. The FRAM has 512 kB of storage available [15]. This allows operations to run without filling the storage as long as the experiment is run once.

3.4.6 Risk Mitigation Matrix

Table 3.4.2 - Risk Mitigation Matrix

Risk	Severity Score	Trend	Mitigation	Validation
001 Collision	Moderate	→	Frequent orbital determination	The satellite demonstrates it will autonomously redirect if others are too close.
002 Follower Coordination Failure	Low	→	Attempt to contact the follower with the ground station.	Ground station was able to contact both satellites.
003 Commander Coordination Failure	Low	↑	Follower contacts ground station so ground can attempt contact with coordination	Ground station was able to contact both satellites.
004 Driver crash	High	↑	The computer will restart in the event of a crash. If the OBC is unable to restart it is reset to the initial settings stored on the flash drive.	Ground tests that demonstrate OBCs ability to restart and reset when a system is intentionally corrupted.

Risk	Severity Score	Trend	Mitigation	Validation
005 Connection failure	Moderate	↓	Additional fastening is included to prevent wires from becoming disconnected from ports and wires are kept within the chassis.	Test the force necessary for disconnection on the ground.
006 Memory failure	Low	↑	If the SDRAM memory system fails the FRAM can be used as the memory system with little difference in delay. If the FRAM has also failed, the Flash may be used as memory with significantly increased delay.	Ground tests where SDRAM is disconnected and operation is able to continue.
007 FRAM failure	Low	↑	The SDRAM may be used to hold data to maintain short write times, but would reduce the memory available. The Flash may also be used to hold data at slower write times and potentially risk corruption of operation critical instructions.	Ground tests where FRAM is disconnected and operation is able to continue.
008 Flash failure	Moderate	↑	More protection is added to the flash storage and the system is initially read-only to prevent any risks caused by writing corrupted data.	Accepted as is

3.5 Power (J. Mitchell)

3.5.1 Power Subsystem Overview

The power subsystem plays a critical role in generating, storing, and distributing electrical energy to sustain the CubeSat throughout its mission. It ensures that all onboard systems and components receive adequate power while maintaining overall stability and operational integrity. The power subsystem manages the efficient allocation of electrical resources to each system based on mission requirements and operational phases, guaranteeing that no subsystem is starved of power at any given time, and preventing any power overload or imbalance that could compromise the CubeSat's functionality.

The power subsystem for the CubeSat mission will utilize two EnduroSat 6U folding solar arrays, mounted on the two 2x3U surfaces of the CubeSat, as its primary source of power generation. Power storage and distribution will be managed by the EnduroSat EPS II system, which integrates an 84 watt-hour lithium-polymer battery. These components were selected to optimize reliability and maximize the likelihood of mission success.

In the CDR phase, a comprehensive power schedule will be conducted to model power generation, consumption, and storage throughout the CubeSat's mission. This is based on the operating mode of the CubeSat and the anticipated power generation and draw for each mode. Using Python, the power level will be computed and visualized on a time scale over a few orbits for verification that that power subsystem will be able to maintain the operation of the CubeSat

3.5.2 Power Schedule

Each component will consume some degree of power on the CubeSat, each with different voltage requirements, and for different periods of time. The average power draw for each component for every CubeSat mode is listed in Table 3.5.1. In the case where specific values could not be provided due to a lack of information provided by the manufacturer, values were chosen that overestimated the actual use of the component to ensure mission success. The columns highlighted in yellow are nominal values that were used. Realistically these values would change according to how heavily the component is used. The payload heater, although highlighted yellow, does change according to the power used as determined in thermal management.

Table 3.5.1 - Subsystem Power Draw by CubeSat Mode [16]

Specific Component Power Draw			Average Power Draw by Mode (W)				
Subsystem	Component	Peak Power (mW)	Tumble	Standby	Experiment	Transfer	Communication
ADCS	Star Tracker (iADCS400)	1000	0	700	1000	700	700
	Reaction Wheels (iADCS400)	15000 (1900)	0-1900	1000	1000-1900	1900	1000
	Magnetorquers, Other (iADCS400)	5000	0	900	900-2000	900	900
	Accelerometer, Gyroscope (IMU Pulse-40)	300	300	300	300	300	300
	GNSS	1000	0	1000	1000	1000	1000
C/D	Computer/Bus	400	400	400	400	400	400
Payload	STARI	20000	0	1200	20000	1200	1200
Power	Solar Panel	85	85	85	85	85	85
Propulsion	Thruster	12000	0	1000	12000	12000	1000
Telecom	Antenna	40	0	40	40	40	40
	Transceiver	1400	0	400	400	400	1400
Thermal	Payload Heater	25600	0-20000	0-20000	0-20000	0-20000	0-20000
	Battery Heater	400	0	400	400	400	400
Totals			785	7425	35625	19325	8425

The most intensive mode by far is experiment mode, as discussed in command and data, will consist of the CubeSat attempting to lock on to the target star and transmit the starlight between the two CubeSats. This requires most of every subsystem besides telecommunications, including the payload itself requiring 20 watts alone.

The table showing the power generation for each mode is listed below. In the table is listed the mode, and the assumption for each mode. The assumption states how the solar panels are deployed, and how the power generation for each mode is calculated. The power generation for each mode is then listed beside each mode.

Table 3.5.2 - Subsystem Power Draw by CubeSat Mode

Mode	Assumption	Power Gen (watts)
Detumble	Solar panels undeployed, weighted power generation among all sides.	6400
Standby	Solar panels deployed, 2 panels facing toward direct sunlight	38400
Experiment	Solar panels deployed, weighted power generation among all sides.	12800
Transfer	Solar panels deployed, weighted power generation among all sides.	12800
Communication	Solar panels deployed, weighted power generation occurs on face of only one solar panel (CubeSat turned 90 degrees relative direction of sun)	19200

From the values presented in the tables, a power schedule for the mission was created over the course of 5 orbits (about 8 hours). This power schedule is presented in Fig 3.5.1. During this power schedule, the power draw reaches a maximum of 46 watts during experiment, and the power level drops to a minimum of 60 watt-hours, again during experiment. It should also be noted that the below power schedule takes into account eclipse period by using a beta angle of zero degrees, which would be the most demanding.

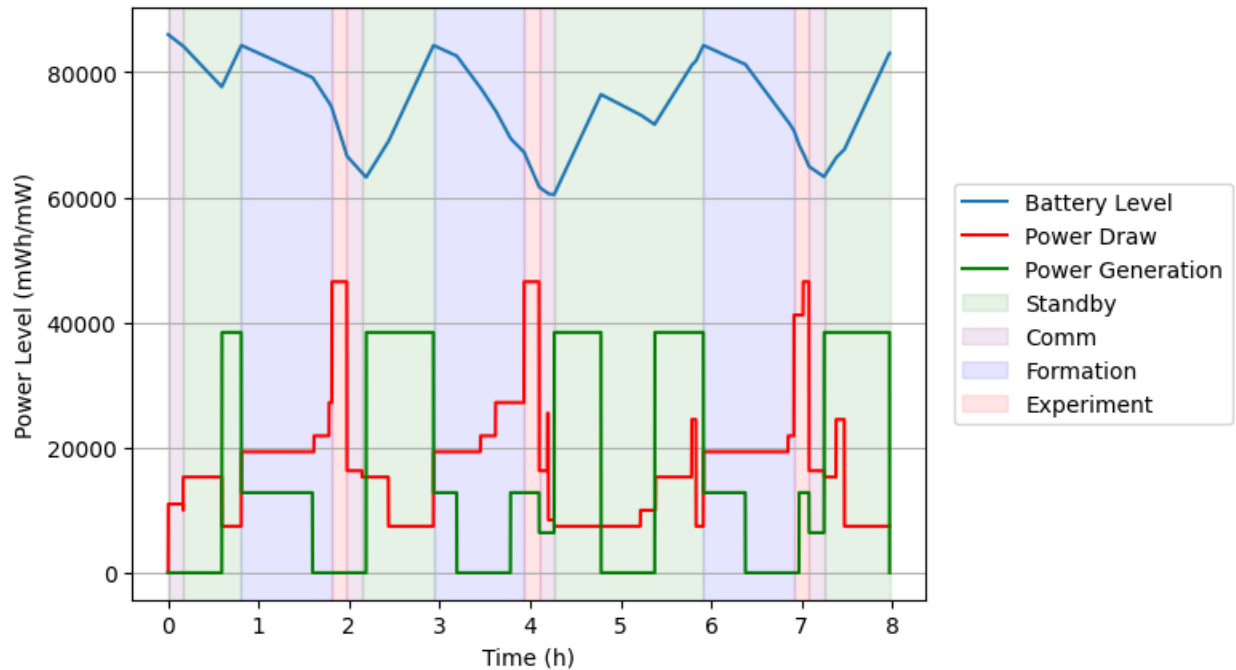


Figure 3.5.1 - Nominal Power Schedule for 5 Orbits

Cycle life can be derived from the power schedule. The battery chosen is rated for up to 500 cycles. From the power schedule, it was determined that approximately 5-6 cycles occur per 24 hours, meaning that over the course of 1 year, approximately 2000 cycles will be needed for the battery.

Because the battery will need significantly more cycles than predicted in the PDR, the battery material which was previously chosen to be lithium-polymer based will be changed for a lithium-iron phosphate material. This is an option the manufacturer provides, and changes the weight very slightly, enough to where it's negligible. This new material can handle up to 5000 cycles as determined within the PDR for this project.

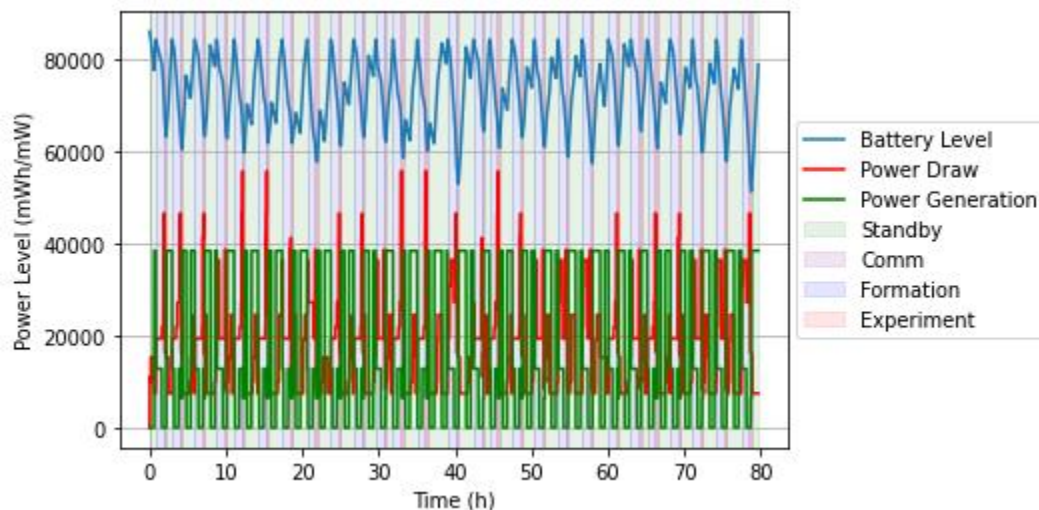


Figure 3.5.1 - Nominal Power Schedule for 50 Orbits

The power schedule was run for a longer period of time of 50 orbits, and it was found that the power level hadn't been in a deficit of more than 34 watt-hour throughout the schedule. This suggests that a smaller battery can be used; however for the sake of the success of the mission, it was decided to stick with the 86 watt-hour battery previously chosen in the PDR phase. This allows for a longer allowable time to detumble, and provides extra redundancy in the power schedule in the circumstances where issues arise.

3.5.3 Risk Assessment & Mitigation

Preparing for risks and developing mitigation strategies will help to boost the success of the mission. There were four main risks identified for this mission and listed below along with there verification strategies:

Table 3.5.3 - Risk Mitigation Matrix

Risk ID/Title	Severity Score	Trend	Mitigation	Verification
001 Experiment Takes Too Long To Lock On and Drains Battery	Moderate	→	At a specified battery threshold, the CubeSat will cease experiment operations and return to standby to reserve power.	Power Schedule Demonstrating Fail Safe Mechanism
002 Detumbling Takes Too Long and Drains Battery	Moderate	→	Limit maximum power draw during tumble to the rate of power generation to ensure no loss of power.	Verify how long CubeSat can remain in tumble using the power schedule and ensure ADCS can work with the power given.
003 Partial Hardware Failure	Moderate	↑	Power level redundancy when acquiring formation/performing experiment; only done when battery is above a threshold.	Power schedule scenario where one or both solar panels fails to deploy
004 Total Hardware Failure	Moderate	↑	Power components with flight heritage only are used – in the scenario it still happens it's accepted.	Accepted as is

The verification for the mitigation strategies described in the risk mitigation are below. Each verification includes a power schedule where the mission risk occurs, and displays how each mitigation strategy would work in the specified scenario.

Shown in the figure below is a scenario where the experiment is unable to lock on, and the experiment automatically turns off.

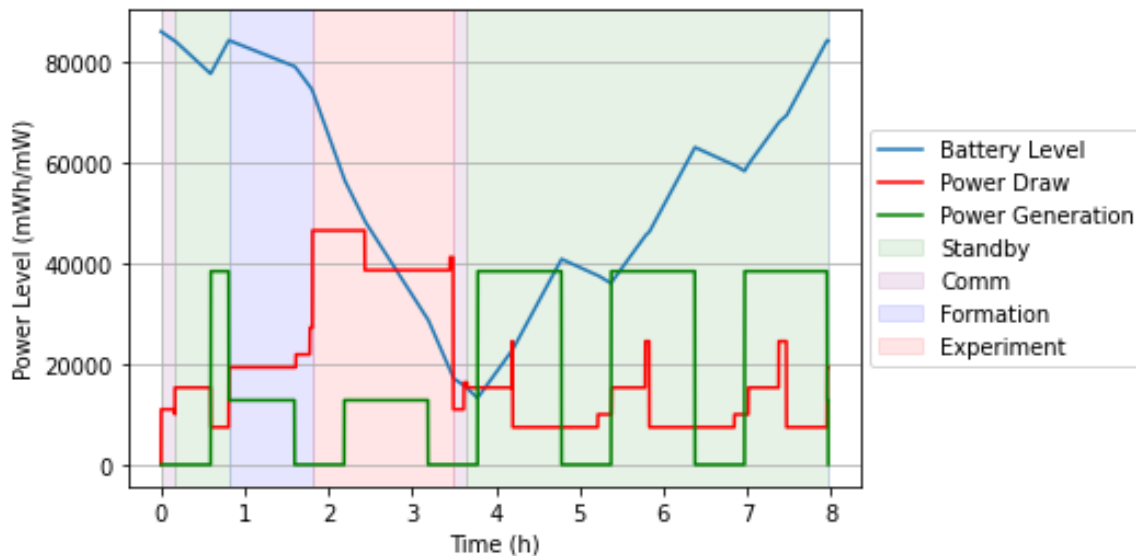


Figure 3.5.2 - Power Schedule Demonstrating Experiment Automatic Shutoff

In the case of the previous figure, the experiment was designed to shut off once the battery reaches 20% battery level.

Tumble is something that is almost guaranteed to happen during the mission. On some missions, detumbling can last as long as a few days, and it's necessary for the CubeSat to survive as long as possible while in tumble. Below is a figure showing the maximum amount of time the CubeSat can remain in tumble while maintaining battery. For the components used for this mission, this maximum time is about 32 hours, which should be more than sufficient enough for detumbling.

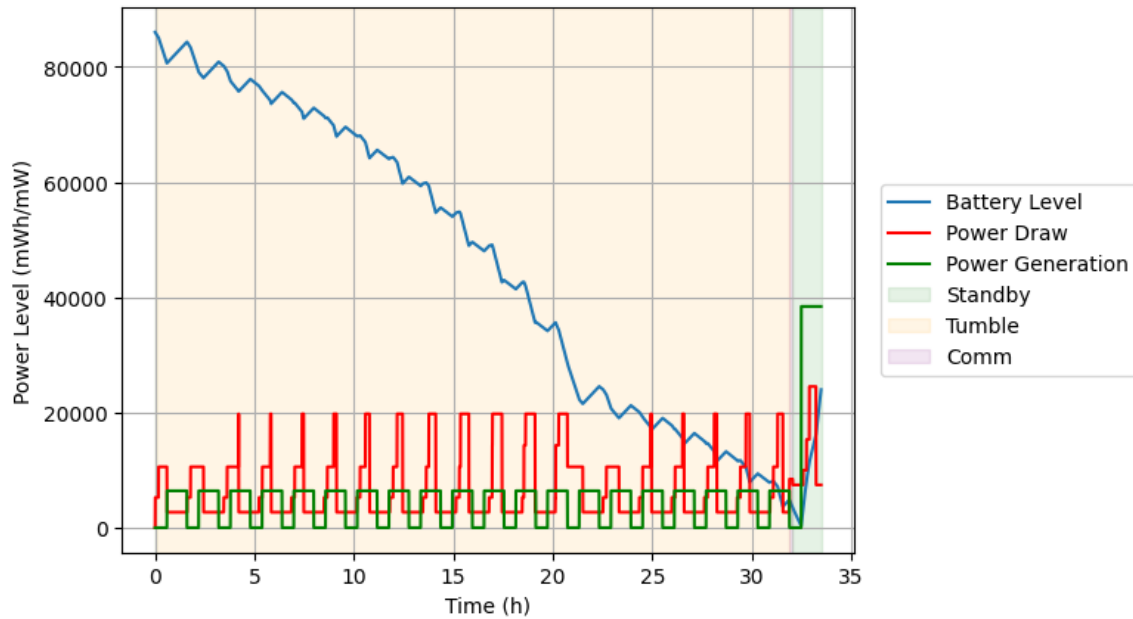


Figure 3.5.3 - Power Schedule with Maximum Allowable Time In Tumble Shown

The last risk identified is the case that one of the solar panels doesn't deploy. In this case, as shown by Fig 3.5.4, the time between experiments is significantly increased, but it can be shown that the experiment can still be carried out, which is critical for mission success. Although the chance of a solar panel not deploying is incredibly rare, it's still important to display.

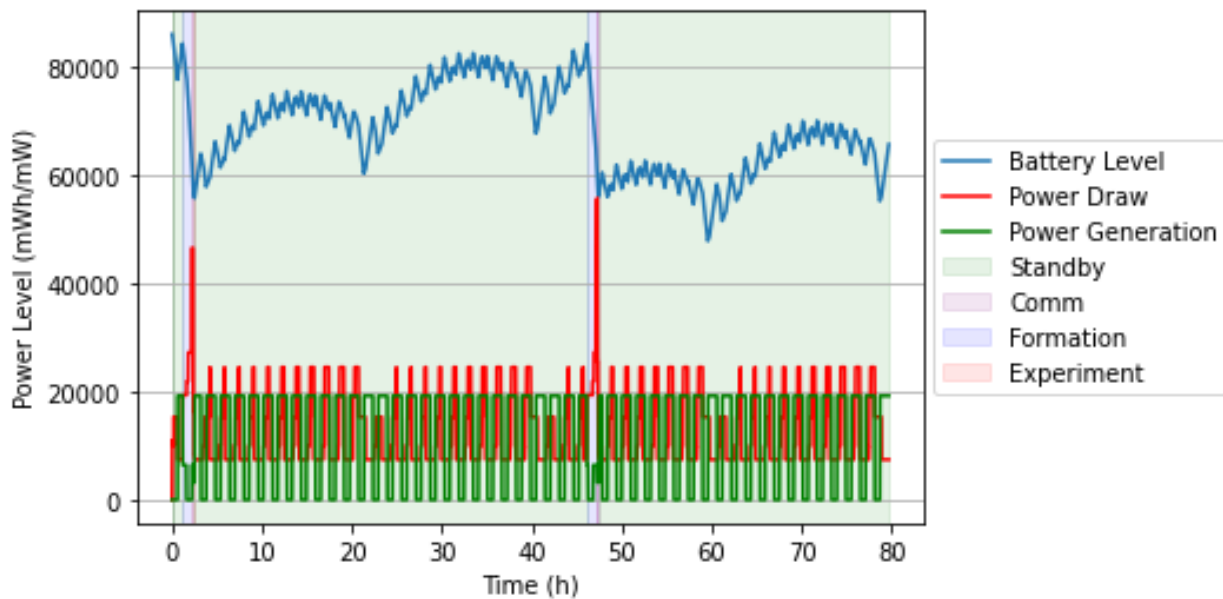


Figure 3.5.4 - Power Schedule with Undeployed Solar Panel

3.6 Telecommunications (A. Cohen)

3.6.1 Telecommunications Analysis Overview

Telecommunications is responsible for the radio communication between the satellite and the receiver on the surface of earth. The ground station will receive payload and engineering data must be sent to the ground for analysis and commands must be able to reach the satellite to adjust operations to accomplish the mission. Communication is only possible during the communication window, where the satellite is in range of the ground station and not blocked by the planet. This window must be calculated based on the range of the transmitter and receiver of the satellite and the ground, the height of orbit, and the speed of the craft. Once the window is calculated, the frequency the orbit will intersect with the window will be determined by a simulation in STK.

The speed of the satellite in orbit relative to the station on the ground will result in a varying distortion in the frequency of the radio signal due to the Doppler Effect. The speed relative to the ground station can be calculated over the duration of the path within the communication window and the resulting distortion can be predicted and compensated for based on an analytical simulation. The communication window will likely be brief, measured in minutes. This demands a large amount of data to be communicated in a short period of time, requiring the Telecommunication system to prioritize certain types of data depending on its necessity for the mission. The data budget is calculated to find the average amount of data that can be transmitted to the ground.

3.6.2 Visibility Simulation

Using the Murgas Transceiver [18] and ISISpace Deployable Antenna [19], the radio communication system has the following properties during operation.

Table 3.6.1 - Hardware Specifications

Frequency	Transmission Power	Receiver Power	Transmitter Gain	Receiver Gain	Loss
145 MHz	27 dBm	-104 dBm	10 dBi	10 dBi	10 dB

The distance of the satellite in orbit that communication is possible is found using the Friis Transmission Equation (Equation 3.6.1). In the equation, R is the range of the signal and λ (lambda) is the wavelength calculated using Equation 3.6.3, both measured in meters (m). P_T is the power of the signal when it is transmitted and P_R is the power of the signal when it reaches the receiver in watts (W). G is the gain of the antenna, with T and R subscripts for the transmitter and receiver respectively. The gain is given in logarithmic decibel units (dB) and must be converted to a linear scale using Equation 3.6.3.

f is the frequency of the signal in hertz (Hz) and c is the speed of the wave, in the case of radio waves the speed would be the speed of light ($3 \cdot 10^8$ m/s). The range was found to be

$3.4 \cdot 10^{25}$ meters, which means the communication will be limited only by interference and not distance.

In order to calculate the portion of the orbit within the communication window the angular portion of the orbit where the ground station was within sight of the satellite was found using Equation 3.6.4. This equation uses the trigonometric identity to calculate the angle (θ) between the ground station at radius R_E and the perpendicular intersection of the orbit at radius R_O . This angle is used with Equation 3.6.5 as well as the angular velocity from Equation 3.6.6 to find the time the satellite will be in the window

Table 3.6.2 - Communication Window

Angular Span	Time
0.8023 rads (36.775°)	9.77 minutes

This calculation assumes a ground station directly below the orbit with a regular period (ideal conditions) and was verified by a simulation in STK. In the simulation, the ground station was assumed to be located at Rensselaer Polytechnic Institute (110 8th St., Troy, NY, 12180). The average time visible in the simulation is 9.67 minutes (580 seconds).

Table 3.6.3 - STK Simulation

Simulation Duration	Average Time	Maximum Visible Time	Minimum Visible Time	Total Visible Time
24 Hours	580.555 s	732.239	445.634 s	3483.331 s

3.6.3 Link Budget

The signal is able to send two bits of binary data each period. At 145 MHz this allows for 290 Mbits/s (36.25 MB/s). Over the average length of the communication window this amounts to 139.2 GB (139,200 MB) of data to be communicated each window. Because the satellite must receive a confirmation that the data has been received by the ground, that requires around half the data to be uplink, limiting the downlink to 70 GB. The total signal loss is found by summing the loss in free space (Equation 3.6.11), the loss from the transmitter (Table 3.6.1), and the loss from the receiver (Table 3.6.1).

Table 3.6.3 - Link Budget

Free Space Loss	Transmitter Loss	Receiver Loss	Total Loss	Budget
-35.6793 dB	10 dB	10 dB	55.6793dB	69.5998

3.6.4 Doppler Effect

The distortion to the frequency is found using Equation 3.6.7 where f_r represents the frequency received and f_t is the frequency when it was transmitted. V is the relative velocity of the craft in orbit and c is the speed of light in a vacuum. The maximum distortion of the frequency allowed for 145 MHz by the FCC is $\pm 0.0005\%$ [21].

The velocity of the CubeSat normal to the ground station was determined by calculating the change in distance over time. The distance of the satellite was found using Equation 3.6.8 with λ as the angle of the orbit centered at the ground station ($\lambda = 0$ when the satellite is directly above ground station). η is the nadir angle, the angle between the target and the ground directly below from the satellite (Equation 3.6.9). ε is the angle from the ground station between the satellite and the line tangent to the surface of the earth, this is called the spacecraft elevation angle. ρ is angle from the satellite between the SSP and the horizon of the Earth (Equation 3.6.10).

Figure 3.6.1 - Radial velocity of the satellite relative to the ground

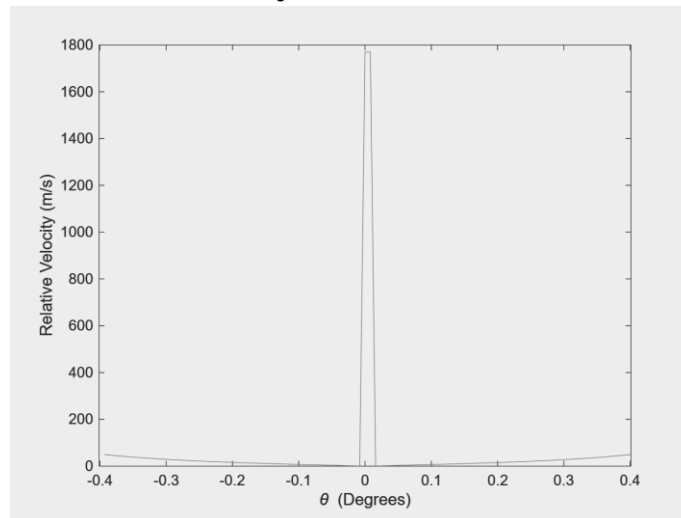
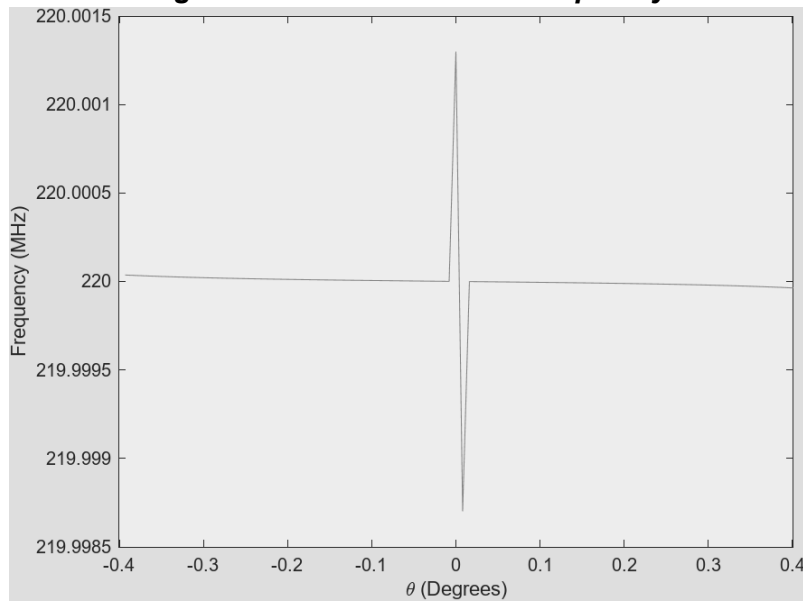


Figure 3.6.2 - Distortion of Frequency



The frequency will be distorted by a maximum of $\pm 0.00059\%$, which will require the satellite to cease transmission for a brief period in order to comply with the FCC.

3.6.5 Inter-Satellite Communication

The satellites will communicate using the onboard radio system at a frequency of 440 MHz. The higher frequency allows for data to be communicated faster at the cost of a reduced penetration and range that is not needed for communication between the satellites flying together. The follower satellite will need to send data such as its position and state to the commander satellite and the commander will need to send commands to the follower.

3.6.6 Risk Mitigation Matrix

Table 3.6.4 - Risk Mitigation Matrix

Risk Title/ID	Severity Score	Trend	Mitigation	Verification
001 Antenna failure	Moderate	↑	The satellite includes an additional antenna, if one fails the other can be oriented towards the target.	Ground tests to ensure both antennae are fully functional.
002 Receiver failure	Low	→	If the receiver on the commander is unable to receive signals from the ground but can receive signals sent from the follower, the satellites will switch roles with the commander becoming the follower and vice versa.	Both satellites are tested in both roles before launch to ensure they are both equally capable in role.
003 Transmitter failure	Low	→	If the receiver on the commander is unable to transmit signals to the ground but can transmit signals sent to the follower, the satellites will switch roles with the commander becoming the follower and vice versa.	Both satellites are tested in both roles before launch to ensure they are both equally capable in role.
004 Doppler Shifting Beyond Range	High	→	The transmission would be shut off when the shifting causes a distortion beyond the legal limit.	Simulations run to find the part of the window beyond the limit

3.7 Propulsion (N. Jankowski)

3.7.1 Propulsion Analysis Overview

In order to ensure the propulsion system is capable of executing successfully on this mission, various required maneuvers must be defined and relative error in these maneuvers must be accounted for. The first analysis will tabulate the errors inherent to the propulsion system, as the accuracy of a maneuver ties directly to the success of an experiment. The second analysis involves using those tabulated values to compute how the satellite will actually respond to the burn. This uses the vehicle's inertial data to calculate the angular acceleration caused by a specific maneuver. This can later be used in an ADCS simulation to determine if the angle control is capable of counteracting the moment caused by the additional force to still achieve the desired final position.

3.7.2 Thrust Error Analysis

In the context of the STARI mission, precise velocity control will be imperative for experiment success. As mentioned before, one satellite will be focused on maintaining orientation to the star of interest, while the other one will be focused on maintaining orientation to the other's transmitter. This means that both satellites will need location, velocity and attitude control from the propulsion and ADCS systems. During an experiment, the propulsion system cannot fire, as that would disturb the transmitting/receiving process. Otherwise, the system is capable of supplying 1 directional DoF and 2 angular DoF.

The propulsion system selected (MiPS) utilizes 4 25 mN cold gas thrusters that can be bolted to the system in a variety of different angles. One of the benefits of this system is it has a pulse-angle range of 0° to 30°. This range accommodates high-impulse burns to take place quickly and allows precise maneuvering during formation flight. Other key parameters of the propulsion system and the satellite as a whole can be seen in Table 3.7.1. It is important to note that the ΔV and response time error are taken from similar systems produced by VACCO Industries, but use identical propellant and have similar thrust levels.

Table 3.7.1 - Propulsion System Characteristics [22][23][24]

Parameter	Value
Spacecraft Mass	9.145 kg
Propellant Mass	0.209 kg
Minimum Impulse Bit	0.05 mN - s
ΔV Error (1- σ)	$\pm 2\%$
Response Time Error (1- σ)	5.08 ± 1.078 ms

To clarify, response time refers to the delay from when the command to open the thruster valve is sent to when the valve actually opens or closes. This impacts the total burn time, which has a direct effect on thrust imparted to the vehicle. Since there is uncertainty for both opening and closing, this error is doubled before adding it to the theoretical burn time length. Equations to determine the thrust certainty can be seen in Appendix B.

Understanding errors inherent to the propulsion and ADCS systems is essential in approximating maneuver accuracy and simulating satellite movement. Table 3.7.2 summarizes the different maneuvers as dictated by flight mechanics (Table 3.2.8) and the error inherent to the thrust.

Table 3.7.2 - Thrust

Maneuver	ΔV Required (m/s)	Burntime (s)	Theoretical Thrust (mN)	Actual Thrust (mN)
Circularization	6.83	650	96.0928	96.1038 ± 5.177
Formation Acquisition	1	100	91.45	91.4638 ± 4.955
Standby	0.0419	10	38.3176	38.2874 ± 2.039
Transfer	0.0284	5	51.9436	51.8414 ± 2.782
Experiment	0.0518	10	47.371	47.2981 ± 2.557

The errors in thrust may or may not be acceptable, requiring further analysis incorporating rotation caused by thruster fire and initial angle error. The torque vector can be found by utilizing the thrust vector and position vector to center of mass, all with respect to the body frame of the satellite.

Table 3.7.3 - Torque

Maneuver	Torque Vector (N * mm)		
		T_x	
		T_y	
		T_z	
Circularization		0 1.97043 ± 0.106 $-0.02072 \pm 1.116E-03$	
Formation Acquisition		0 1.87529 ± 0.102 $-0.019719 \pm 1.07E-03$	
Standby		0 0.78501 ± 0.0418 $-8.2548E-03 \pm 8.69E-03$	
Transfer		0 1.0629 ± 0.0571 $-0.0112 \pm 5.77E-04$	
Experiment		0 0.9713 ± 0.0523 $-0.01021 \pm 5.55E-04$	

With the torque caused by propulsion understood, the angular acceleration caused by thruster fire can be calculated using the (deployed) moment of inertia of the vehicle. These angular accelerations will be used by the ADCS simulations to determine if the design is capable of

controlling the attitude well enough to complete the various maneuvers required. Table 3.7.4 displays these values. Note that the upper limit of the torque vector was taken to ensure that ADCS can handle the most extreme angular accelerations caused by propulsion.

Table 3.7.4 - Angular Acceleration

Maneuver	Angular Acceleration Vector (rad/s ²) [α_x , α_y , α_z]
Circularization	[-5.18E-05, 0.01742, -2.34E-04]
Formation Acquisition	[-4.938E-05, 0.01659, 2.23E-04]
Standby	[-2.062E-05, 6.939E-03, -9.313E-05]
Transfer	[-2.793E-05, 9.40E-03, -1.262E-04]
Experiment	[-2.55E-05, 8.579E-03, -1.151E-04]

3.7.3 Risk Mitigation Matrix

A major source of risk in the propulsion system stems from reporting what the system actually achieved. As shown in Section 3.7.2, the error in thrust (magnitude and direction) can likely be accounted for to a certain degree by coordinating with the ADCS system. However, when imparting an impulse on the satellites, the filter utilized by the ADCS system may not account for the change in velocity soon enough, causing a large error to appear. This large jump in sensor data needs to be autonomously predicted so that the filter is aware of the instantaneous change in velocity.

Other sources of risk are characteristic of the propulsion system itself. Since a cold gas system is being used, the stored, pressurized gas must be kept at an appropriate temperature to ensure engine efficiency. With too cold propellant, the engine may impart unsteady thrust during short burns (chugging), leading to an inaccurate maneuver. Valve failure, pressure leaks, or fire mistiming are all other potential sources of risk that are found in every propulsion system.

Table 3.7.2 - Risk Mitigation Matrix

Risk ID/Title	Severity Score	Trend	Mitigation	Verification
001 Sensor Jump	Moderate	→	Filters used by the ADCS system will be aware of instantaneous burns to avoid large errors as it catches up to the new velocity profile.	Simulations can be run where the ADCS system is fed planned changes in velocity and direction to simulate a burn, and errors of various cases can be studied.
002 Nonoptimal Firing	Low	→	Equip the propulsion system with a back-up heater and ensure that the system is only used when it is within its operating range.	Collect data at various temperatures to ensure the vendor is communicating the actual thermal limits of the unit and study the changes in thrust error.
003 Lack of Propellant	Moderate	↑	The ΔV has margin built in to ensure this risk is mitigated. Propellant use can be scrutinized throughout the mission by only attempting formation flight when the required propellant amount is at a suitable level.	Perform leak tests on the propulsion unit to ensure that the leak rate will not jeopardize the mission timeline.
004 Thruster Fails to Fire	Low	→	Enstate a checking system for successful fire. In the event of an unsuccessful fire, recheck if burn conditions are still met before firing again.	Do ground tests with dummy commands to ensure that the system rechecks for burn conditions.

3.8 Mechanisms & Deployables (J. Mitchell)

3.8.1 Mechanisms & Deployables Subsystem Overview

The mechanisms & deployables subsystem deals with any fixtures that may need to be “deployed” from the payload, that is to say they will extend outwards from the surface of the CubeSat after deployment from the LV. This can include antennas, solar arrays, cameras/sensors, booms/solar sails, telescopes and more. The CubeSat in this mission will only have three main deployables, the solar panel array, deployable antenna, and the drag sail. As the CubeSat deployer is often considered in the deployables subsystem, this will be discussed also. The specific components that were chosen are as follows: the Exo Launch EXOPOD Nova 12U S2 CubeSat deployer, the ADEO PICO drag sail, ISIS Space deployable antenna system, and the two EnduroSat 6U folding solar arrays.

For the CDR phase, a deployment schedule detailing the deployment times for deployables onboard the CubeSats including the solar arrays, antennae, and drag sails will be discussed.

3.8.2 Deployment Schedule

A well-structured deployment schedule is vital for the mission's success, ensuring that each component is deployed at the appropriate time to optimize functionality and minimize risks.

For the deployment of components, it is assumed that the CubeSat is detumbled. Components, while they can be deployed in tumble, are usually deployed when the CubeSat is detumbled as it will affect the inertia of the satellite, and may lead to malfunctions in the deployment that are unwanted.

The CubeSat, while not technically being deployable, is included in the deployment schedule. This gives a baseline for which to base other components timings off of. The time at which the CubeSat is deployed from the launch vehicle via the deployer is not exactly known, and would not be known until before the mission launches. Because this doesn't have any impact on the mission anyway, this is left as to be determined.

Solar panels need to be deployed as soon as possible. That being said, there are guidelines set by both CubeSat standard and launch vehicle standard that must be abided by. SpaceX requires deployment of components to wait >120 seconds. However, the CubeSat design guide says there must be at least 30 mins between the deployment from the launch vehicle and the deployment of components. To be safe, this is set to 30 minutes. [29] [30]

The antenna will be deployed nearly directly after the solar panels are deployed. A 5 minute buffer between the deployments is given to ensure that no unwanted interactions occur between the two.

The drag sail will only be deployed after the mission is concluded; concluded meaning that either the mission has successfully demonstrated the goal it set out to achieve, it has partially demonstrated the set goal but is unable to fulfill the other goals, or the mission is deemed a failure. In the case that the mission fails to make contact with the earth, and the mission cannot be carried out, the drag sail will have a default time of 1 year to deploy. This will serve to deorbit the satellite to ensure all regulations are met.

The estimated operation time for each component to deploy is included alongside each deployment. These operation times were determined from the specification sheets given by the manufacturer, and by looking at other missions involving CubeSats.

Table 3.8.1 - Deployment Schedule

#	Deployable	Time to Deploy	Estimated Operation Time
1	CubeSat Deployer [25]	TBD by LV	<20 seconds
2	Solar Panels [28]	>30 minutes after CubeSat deployment	<5 seconds
3	Antenna [27]	>5 minutes after solar panel deployment	<1 minute
4	Drag Sail [26]	<1 year or until mission success	<1 minute

3.8.3 Risk Assessment & Mitigation

This risk mitigation matrix outlines potential failure modes associated with the deployment schedule of the CubeSat mission, assigns severity scores to each risk, and identifies the necessary mitigation and verification steps to minimize their impact. By addressing critical risks such as deployment mechanism failures for the launch vehicle (LV), solar panels, and antenna, as well as the drag sail deployment, this table ensures that all key components can be deployed successfully. Each mitigation strategy involves testing and redundancy measures, while verification steps are implemented to confirm the reliability of these mechanisms prior to launch. This approach enhances mission success and minimizes potential operational and regulatory setbacks within the scope of the mission.

Table 3.8.2 - Risk Mitigation Matrix

Risk ID/Title	Severity Score	Trend	Mitigation	Verification
001 LV Deployment Failure	Moderate	→	Intensive ground testing to ensure electro-mechanical latch mechanism is operational, and contact with rails is adequate.	Conduct mechanical vibration and thermal testing under simulated launch conditions to verify latch mechanism functionality. This is carried out by SpaceX, and will be included within the contract with SpaceX.
002 Solar Panel, Antenna Deployment Failure	Moderate	→	Solar panels have triple redundant burn wire, antenna has dual redundant burn wire.	Perform pre-flight deployment tests for burn wires and document their success rates.
003 Drag Sail Deployment Failure	Moderate	→	Accepted; no major impact on mission success, but would result in fines for not deorbiting satellites.	Accepted as is

3.9 Thermal Management (N. Jankowski)

3.9.1 Thermal Analysis Overview

In the scope of this report, the thermal system has fairly limited simulation work to do. In reality, a full thermal analysis should be conducted, investigating localized thermal loads and heat generation of each PCB, defining the factor of safety of each component. However, at this time, only the average satellite temperature will be of concern as it completes an orbit and performs necessary functions. To do this analysis, a script will be created that tracks various characteristics of the satellite with relation to time. One of these characteristics is the environmental heat environment that the spacecraft finds itself in. Eclipse period impacts the solar constant, beta angle of orbit impacts solar irradiance, and the emissivity or absorptivity of each surface impacts this calculation. Another characteristic is the heat generated, which will require a detailed time-based power schedule to be known. Combining these two characteristics allows the unaltered, average temperature of the satellite to be found, which can highlight specific times where the heaters need to be activated.

3.9.2 Thermal Environment Schedule

Appendix D summarizes the equations and values used to determine the average temperature of the satellite throughout its orbit. During this analysis, 4 major assumptions were made, as detailed below.

1. For both solar panels and painted surfaces, only $\frac{1}{2}$ of its total surface area is exposed to a thermal environment. This assumption is valid as in no configuration will the satellite be exposed to solar or infrared energy on all surfaces.
2. The emissivity and absorptivity values were assumed to be 0.80 and 0.10 respectively. Vendor specifications sheets do not outline this parameter, so a reasonable estimate was used.
3. The specific heat of the satellite was assumed to match that of aluminum. Each component is composed of various materials, and specific heat is not a common value to report. In reality, the specific heat of the system is likely to be much lower.
4. The beta value of the orbit was assumed to be 0, 45, 70, or 90 degrees to match the albedo information available. This range captures what the satellite will realistically see across a long mission duration.

Table 3.9.1 summarizes the operating range of each component. Specifically, the target is to stay between -10°C and 40°C , giving the satellite comfortable thermal margin on either side. These constraints are vital when determining the threshold to activate the heaters to ensure that no component dips below its minimum value at any point.

Table 3.9.1 - Component Thermal Operating Range

Component	Minimum (°C)	Maximum (°C)
Payload	-40	90
Structure	-40	80
Propulsion	-15	55
Command & Data	-25	65
Telecommunications	-20	60
Power	-20	60
ADCS	-20	60
Orbit Temperature	-65	125

As a control case, the satellite temperature with a base power schedule (Figure 3.5.1) and no heater activation was plotted. This is done to highlight the need for heaters, specifically during the eclipse period. This eclipse period lasts 37% of the total orbit time and offers much lower environmental heating, meaning the temperature is largely dependent on internal heat generation. However, because the power system reduces power consumption during times of no sunlight (such as eclipse), this places an even larger importance on proper heater usage. The plot of temperature versus time for 10 orbits can be seen in Figure 3.9.1.

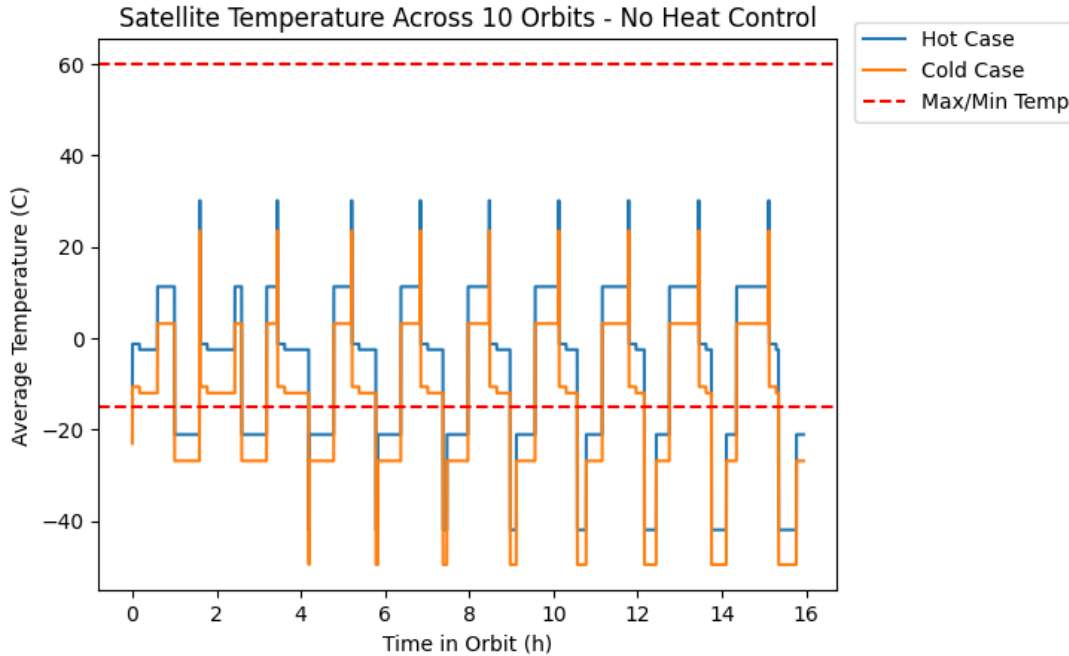


Figure 3.9.1 - No Heat Control Plot

As it can be seen, the average temperature of the satellite dips below the minimum temperature allowed (-10°C), representing a potential risk to thermally-sensitive components. Note that Figure 3.9.1 uses a beta angle of 0° , which offers the least amount of albedo to the satellite. This was done to ensure that the worst-case thermal scenario was captured. Next, a simple check for heater input was added. Once the temperature dips below the minimum target, the satellite will determine how much power the heaters require to raise the satellite temperature back to the safe zone across the duration of the eclipse period. This can be seen in Figure 3.9.2.

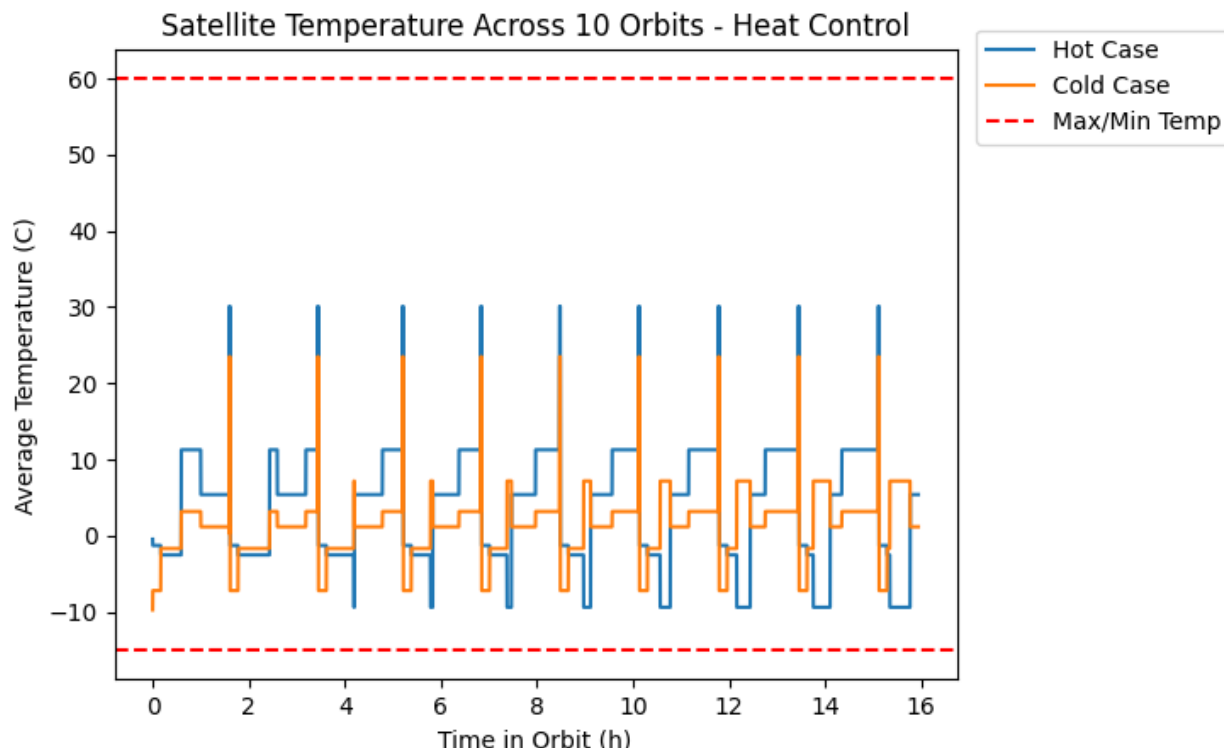


Figure 3.9.2 - Heat Control Plot

Now, in both hot and cold cases, the satellite temperature sits comfortably within the thermal envelope. As the power schedule becomes more refined, the peaks will flatten out as large jumps in wattage usage will be lowered. Note that this plot also assumes a beta angle of 0° and a maximum heater wattage of 20W to avoid burn-out (the selected heater has a maximum usage of 25.6W).

3.9.3 Risk Mitigation Matrix

The thermal subsystem is critical but fairly robust for this mission, largely because of the simple mechanics used to heat and cool the satellite. The most critical risk involved in this system is its ability to keep the satellite within its thermal envelope during tumble. Because the length of tumble is unknown, there needs to be substantial margin built into the power schedule to sustain the heaters during eclipse mode. As it stands, the satellite will be able to survive for ~ 32 hours in tumble mode before satellite death. There are potential risks present for the coating and heaters used, but they are limited. The AZ-93 coating may experience degradation in its potential

emissivity and absorptivity over time as the paint wears off or chips. This could lead to more extreme temperature fluctuations, which may push the satellite outside of the operational margin. The HAP-6743 heaters are also reliable and should have extensive testing before launch. However, there is always the possibility of heater failure, especially over the course of a 2 year mission. Similarly, there needs to be a method to shut down a heater in the event of an autonomous failure. In both cases, multiple heaters can be used for redundancy and due to their low cost, mass, and power requirements, would be simple to integrate to the system.

Table 3.9.2 - Risk Mitigation Matrix

Risk ID/Title	Severity Score	Trend	Mitigation	Verification
001 Heater Failure	Low	↑	Design for redundant heaters that can take over in the event of one heater failure.	Test each heater's functionality before flight and the satellite's ability to recognize heater failure and enable back-up.
002 Tumble Mode Lasts Too Long	High	↑	Reduce detumble power usage as much as possible and ensure heaters are installed on the most thermally-sensitive components	Test the satellite's detumble mode before flight, tracking power usage. Power schedule and thermal schedule scripts can be used to verify time of life.
003 Paint Degradation	Low	↑	Extensive testing led by NASA shows minimal thermal performance degradation over time	Accepted as is
004 Heater Stuck On	Moderate	→	Manual bypass to shut off heater and switch to back-up.	Ensure that the bypass works properly. In the case that uplink is not possible with the satellite, the thermal simulation was rerun with a heater constantly on. The satellite was still within the thermal envelope, and may just be power-starved until the command can be sent.

4 Summary & Conclusion (J. Mitchell)

The CDR phase of the STARI mission has established a robust framework for achieving mission objectives by thoroughly validating subsystem designs through simulations, schedules, and risk mitigation strategies. Each subsystem has been scrutinized to ensure its readiness to meet the technical, operational, and safety requirements of this groundbreaking CubeSat mission. This phase not only reinforces the feasibility of STARI's objectives but also provides the confidence needed to transition into the implementation phase.

The largest success in the structure subsystem was creating the first iteration CAD model, proving that the selected components are capable of fitting within a 6U chassis and providing required technical data to other subsystems. Flight mechanics subsystems utilized Clohessy-Wiltshire equations to create a simulation of the relative motion of the chaser to ensure that the CubeSats maintain the required inter-satellite distances of 10 to 100 meters. Furthermore STK was utilized to obtain a more accurate transfer from standby to experiment along with the beta angle, eclipse period, and decommissioning time. The attitude determination and control subsystem (ADCS) was validated through simulations by using sensor errors and drifts to create a closed-loop system. An extended Kalman Filter was used to reduce the noise of the system and PID control was made to model the actuators to ensure the spacecraft can meet the accuracy needed for the mission. The command and data subsystem created a detailed plan for how the satellites would operate and communicate autonomously to achieve all mission goals. Power used comprehensive scheduling to balance power generation, storage, and consumption, accounting for different operational modes and eclipse periods, confirming that the chosen components would be sufficient for the mission's demands by modeling the power storage through several orbits. Telecommunications simulated and analyzed the radio communication systems on board the satellite and on the surface. Doppler shift was investigated to ensure compliance with federal and international law. Inter-satellite communication was analyzed to ensure reliable communication and coordination. Propulsion system succeeded in verifying the error inherent to the chosen engine and determining the imposed torque to the spacecraft. Further position, angle, and velocity simulations will better qualify the engine's performance. Mechanisms and deployables presented a deployment schedule detailing the times at which components should be deployed. The thermal management succeeded in estimating the average temperature of the satellite throughout its orbit, using a 1-experiment power schedule. It also verified that the heaters are capable of producing enough wattage to the craft to keep it within its thermal envelope.

Through these rigorous analyses and designs, the CDR phase has not only confirmed the mission's technical feasibility but also laid a solid foundation for its successful execution. The integration of simulations, schedules, and risk mitigation strategies across all subsystems demonstrates a cohesive and forward-thinking approach. STARI is now poised to achieve its ambitious goals, advancing space-based interferometry and setting a new standard for CubeSat missions.

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6 Appendices

Appendix A - Structures

Structures Likelihood Consequence Matrix

		Consequence				
		1 Negligible	2 Minor	3 Moderate	4 Major	5 Catastrophic
┌	5 Almost Certain	Low	Moderate	High	High	High
	4 Likely	Low	Moderate	Moderate	High	High
	3 Possible	Low	Low	Moderate	Moderate	High 001
	2 Unlikely	Low	Low	Moderate	Moderate	Moderate
	1 Rare	Low	Low	Low 002	Low	Moderate

Appendix B - Propulsion

$$T = \frac{\Delta v_{sat} * m_{sat}}{\Delta t} \quad (\text{Thrust Equation}) [23]$$

Tposition_vec: [-104.65 , 0, -1.325]

CoM_vec: [11.5778, 0.2156, 19.1781]

T_vec: [T, 0, 0]

r_vec = CoM_vec - Tposition_vec: [116.23, 0.2156, 20.50]

Torque Vector = r_vec x T_vec

Angular Acceleration Vector = Torque Vector x inv([Inertia Matrix])

Propulsion Likelihood Consequence Matrix

		Consequence				
		1 Negligible	2 Minor	3 Moderate	4 Major	5 Catastrophic
┌	5 Almost Certain	Low	Moderate	High	High	High
	4 Likely	Low	Moderate	Moderate	High	High
	3 Possible	Low	Low 004	Moderate	Moderate	High
	2 Unlikely	Low	Low	Moderate	Moderate 001	Moderate
	1 Rare	Low	Low	Low 002	Low	Moderate 003

Appendix C - ADCS

ADCS Likelihood Consequence Matrix

		Consequence				
		1 Negligible	2 Minor	3 Moderate	4 Major	5 Catastrophic
└	5 Almost Certain	Low	Moderate 003 002	High	High	High
	4 Likely	Low	Moderate	Moderate	High 001	High
	3 Possible	Low	Low 004	Moderate	Moderate	High
	2 Unlikely	Low	Low	Moderate	Moderate	Moderate
	1 Rare	Low	Low	Low	Low	Moderate

(Equations used for Euler Angles)

$$roll = u = \tan^{-1} \left(\frac{2(q_0 q_1 + q_2 q_3)}{q_0^2 - q_1^2 - q_2^2 + q_3^2} \right) = \text{atan2}[2(q_0 q_1 + q_2 q_3), q_0^2 - q_1^2 - q_2^2 + q_3^2]$$

$$pitch = v = \sin^{-1}(2(q_0 q_2 - q_1 q_3)) = \text{asin}[2(q_0 q_2 - q_1 q_3)]$$

$$yaw = w = \tan^{-1} \left(\frac{2(q_0 q_3 + q_1 q_2)}{q_0^2 + q_1^2 - q_2^2 - q_3^2} \right) = \text{atan2}[2(q_0 q_3 + q_1 q_2), q_0^2 + q_1^2 - q_2^2 - q_3^2]$$

(Gimbal lock parameters)

<i>If pitch = $+\frac{\pi}{2}$ (radians)</i>	<i>If pitch = $-\frac{\pi}{2}$ (radians)</i>
$roll = 0$ $yaw = -2\text{atan2}(q_1, q_0)$	$roll = 0$ $yaw = 2\text{atan2}(q_1, q_0)$

(Jacobian Matrix for EKF)

$$\mathbf{J}_h \equiv \begin{bmatrix} \frac{\partial h_1}{\partial x_1} & \frac{\partial h_1}{\partial x_2} & \dots & \frac{\partial h_1}{\partial x_n} \\ \vdots & & \ddots & \vdots \\ \frac{\partial h_m}{\partial x_1} & \frac{\partial h_m}{\partial x_2} & \dots & \frac{\partial h_m}{\partial x_n} \end{bmatrix}$$

(Equations used for Iterated EKF)

$$\mathbf{x}_{k,i}^a \approx \mathbf{x}_k^f + \mathbf{K}_k(\mathbf{z}_k - \mathbf{h}(\mathbf{x}_{k,i}^a))$$

$$\mathbf{K}_{k,i} = \mathbf{P}_k^f \mathbf{J}_h^T(\hat{\mathbf{x}}_{k,i}) \left(\mathbf{J}_h(\mathbf{x}_{k,i}^a) \mathbf{P}_k^f \mathbf{J}_h^T(\mathbf{x}_{k,i}^a) + \mathbf{R}_k \right)^{-1}$$

$$\mathbf{P}_{k,i} = (\mathbf{I} - \mathbf{K}_{k,i} \mathbf{J}_h(\mathbf{x}_{k,i}^a)) \mathbf{P}_k^f$$

Appendix D - Thermal Management Calculations

$$Q_{env} = \sum_1^n \alpha_n S [A_{Pn} + R A_{Rn}] + \sum_1^n \varepsilon_n I_R A_{Rn} \quad (\text{Environmental Heat})$$

[31]

$$Q_{env} + Q_{in} = \sigma T^4 \sum_1^n \varepsilon_n A_n \quad (\text{Internal Heat Generation}) \quad [31]$$

Variable	Description	Nominal Value
α	Absorptivity of surface. Fraction of solar radiation absorbed.	AZ-93 Paint = 0.15 Solar Panels = 0.1
S	Solar irradiance on Earth. Energy supplied by the Sun. Also known as the Solar Constant.	1317 W/m^2 (Cold) 1419 W/m^2 (Hot)
A_p	Projected area towards the sun ($\frac{1}{2}$ total surface area).	Painted = 0.09 m^2 Solar Panels = 0.12 m^2
R	Percentage of solar irradiance reflected from Earth (albedo).	0.24 (Cold) 0.34 (Hot)
A_R	Area exposed to diffusely reflected solar energy from Earth ($\frac{1}{2}$ total surface area).	Painted = 0.09 m^2 Solar Panels = 0.12 m^2
ε	Emissivity of surface. How well a surface emits radiation compared to a blackbody.	AZ-93 Paint = 0.91 Solar Panels = 0.80
I_R	Irradiance of infrared energy from Earth.	217 W/m^2 (Cold) 261 W/m^2 (Hot)
A_{IR}	Area exposed to infrared energy emitted from Earth ($\frac{1}{2}$ total surface area).	Painted = 0.09 m^2 Solar Panels = 0.12 m^2
Q_{in}	Internal heat generation, in watts.	Varies based on power schedule
σ	Stefan-Boltzmann constant	$5.67051 \times 10^{-8} W/m^2 K^4$
T	Average temperature of the satellite in Kelvin	Varies based on inputs.

Heat Equation Variables [31]

$$P = \frac{mc_p \Delta T}{\Delta t} \quad (\text{Heater Power Requirements}) \quad [32]$$

Thermal Management Likelihood Consequence Matrix

	Consequence
--	-------------

		1 Negligible	2 Minor	3 Moderate	4 Major	5 Catastrophic
└	5 Almost Certain	Low	Moderate	High	High	High
	4 Likely	Low	Moderate	Moderate	High	High
	3 Possible	Low	Low	Moderate 004	Moderate	High 002
	2 Unlikely	Low	Low	Moderate	Moderate	Moderate
	1 Rare	Low 003	Low	Low	Low 001	Moderate

Appendix E - Flight Mechanics

Flight Mechanics Likelihood Consequence Matrix

		Consequence				
		1 Negligible	2 Minor	3 Moderate	4 Major	5 Catastrophic
└	5 Almost Certain	Low	Moderate	High	High	High
	4 Likely	Low	Moderate	Moderate 003	High	High
	3 Possible	Low	Low	Moderate	Moderate	High 001
	2 Unlikely	Low	Low	Moderate 002	Moderate 004	Moderate
	1 Rare	Low	Low	Low	Low	Moderate

Appendix F - Power

Power Likelihood Consequence Matrix

		Consequence				
		1 Negligible	2 Minor	3 Moderate	4 Major	5 Catastrophic
└	5 Almost Certain	Low	Moderate	High	High	High
	4 Likely	Low	Moderate	Moderate	High	High
	3 Possible	Low	Low	Moderate 003	Moderate	High
	2 Unlikely	Low	Low	Moderate	Moderate 001, 002	Moderate
	1 Rare	Low	Low	Low	Low	Moderate 004

Appendix G - Mechanisms & Deployables

Mechanisms & Deployables Likelihood Consequence Matrix

		Consequence				
		1 Negligible	2 Minor	3 Moderate	4 Major	5 Catastrophic
└	5 Almost Certain	Low	Moderate	High	High	High
	4 Likely	Low	Moderate	Moderate	High	High
	3 Possible	Low	Low	Moderate	Moderate	High
	2 Unlikely	Low	Low 003	Moderate 002	Moderate	Moderate
	1 Rare	Low	Low	Low	Low	Moderate 001

Appendix H - Commands & Data

Commands & Data Likelihood Consequence Matrix

		Consequence				
		1 Negligible	2 Minor	3 Moderate	4 Major	5 Catastrophic
	5 Almost Certain	Low	Moderate	High 004	High	High
	4 Likely	Low	Moderate	Moderate	High	High
	3 Possible	Low	Low 006	Moderate	Moderate 001	High
	2 Unlikely	Low	Low 002, 003	Moderate	Moderate	Moderate 008
	1 Rare	Low	Low	Low 007	Low	Moderate 005

Appendix I - Telecommunications

Telecommunications Likelihood Consequence Matrix

		Consequence				
		1 Negligible	2 Minor	3 Moderate	4 Major	5 Catastrophic
	5 Almost Certain	Low	Moderate	High 004	High	High
	4 Likely	Low	Moderate 001	Moderate	High	High
	3 Possible	Low	Low 002	Moderate	Moderate	High
	2 Unlikely	Low	Low 003	Moderate	Moderate	Moderate
	1 Rare	Low	Low	Low	Low	Moderate

Equation 3.6.1

$$R = \frac{P_T G_T G_R \lambda^2}{16\pi^2 P_R}$$

Equation 3.6.2

$$\lambda = \frac{c}{f}$$

Equation 3.6.3

$$G = 10^{\frac{dB}{10}}$$

Equation 3.6.4

$$\cos(\theta) = \frac{R_O}{R_E}$$

Equation 3.6.5

$$\theta = \omega t$$

Equation 3.6.6

$$\omega = \frac{V}{R_O}$$

Equation 3.6.7

$$f_r = f_t \left(1 \pm \frac{V}{c}\right)$$

Equation 3.6.8

$$D = R_E \frac{\sin(\lambda)}{\sin(\eta)} [20]$$

Equation 3.6.9

$$\sin(\eta) = \cos(\varepsilon) \sin(\rho) [20]$$

Equation 3.6.10

$$\sin(\rho) = \frac{R_E}{R_O} [20]$$

Equation 3.6.11

$$L_S = 20 \log_{10} \left(4\pi \frac{D}{\lambda}\right)$$